A comparative investigation into Cloud-based Active Directory solutions for SMB’s.

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Abstract

This research question was answered by finding the cost of an on-premises server and calculating the setup costs, licensing costs and energy consumption of the server using a custom-built Python tool. Three market leading Public Cloud providers were then compared for their cost and feature benefits between each other and then compared with the on-premises solution itself.

The investigation finds that in every case, utilising a Cloud-based Active Directory management solution is much more cost-efficient and feature rich than building and maintaining the on-premises server infrastructure required to host such a solution.

At the time of the investigation, it was also found that Amazon Web Services was the most cost-efficient Public Cloud platform to run a Windows Server solution on, though Public Cloud provider costs can fluctuate constantly and also be negotiated with the vendor themselves and therefore is not concrete evidence.

For the purposes of the investigation, any one of the investigated Public Cloud platforms is suitable for hosting an Active Directory management solution and should be dealt with on a case-by-case basis by the SMB negotiating with a provider.

In conclusion, the main research question was answered in that Cloud-based Active Directory management solutions are better off built and maintained in the Cloud and that there are various Public Cloud providers with similar features and costs that can host this workload.

Chapter 1 - Introduction

This section introduces the reader of the paper to the topic at hand and what the aims and objectives of the research are. It looks at each investigated area of the topic, such as on-premises servers and Cloud vendors as well as four questions that should be answered in order to receive a quality output in reference to answering the research question.

In order to manage and access resources in the modern world, an attribute needs to be assigned to a user or entity. This is done in the form of user or entity logins or 'identities', which moderate the use of resources and are assigned privileges from an organisational administrator. In an on-premise environment, most identities are managed from a directory solution such as Windows Server 2016 with Active Directory or other comparable solutions.

At the bottom of the research paper an appendices section and a setup guide for Windows Server 2016 Essentials can be found. Throughout the research paper there are figures that do not fit entirely onto a page and are therefore cropped, so it may be necessary to check the appendices for the full base image if required. The setup guide for Windows Server intends to walk the reader through setting up their own on-premises deployment in which they can utilise the organisational chart provided in the research paper or use their own to build their on-premises solution.

Aims and Objectives

This report investigates the relatively new concept of Cloud-based Active Directory management. It is possible now to create and manage user and organisational identities entirely from and in a Cloud environment without needing to own any hardware or configure the software and infrastructure required to create and manage identities from an organisations' physical assets. I will investigate four core areas of Cloud Active Directory management in relation to SMB's (Small-Medium enterprises).

1. How cost-effective is a Cloud Active Directory management solution compared to on-premises Active Directory management solutions?
2. What are the differences in features and capability between an on-premises Active Directory management system and a Cloud-based Active Directory management system?
3. What are the differences in features and capabilities between three market leading Cloud providers for Active Directory management?
4. What are the possible risks and additional risk factors when using the Public Cloud for Active Directory services?

SuperThreads Sheff are an SMB that deal in clothing based in Sheffield, England. In order to manage their digital identities, devices and documents they are using on-premises servers. These on-premises servers must be maintained and constantly powered on as well as being locked away and secured in a physical space.

After hearing about Cloud offerings, the IT Manager at SuperThreads Sheff wants to look into how moving to the Cloud could resolve three of their issues; How can SuperThreads Sheff save money by hosting their Active Directory solution in the Cloud?, What new capabilities and features can a Cloud directory management solution provide?, What are the options available for Cloud-based Active Directory management solutions?

An aspect highly important to the IT Manager is that any implemented solutions also meet compliance needs. This means that the solution must abide by regional compliance regulations, in this case the GDPR.

As well as this, each of the investigated solutions have different capabilities and features. The IT Manager wants to know which of the solutions will best fit their requirements for migrating and managing their Active Directory services in the Cloud – cost effectively, efficiently and with minimal administrative overheads.

This research will give insight into how SMB's can migrate to or adopt Cloud-based directory solutions based on information about the benefits of Cloud Active Directory management solutions. Various market leading Public Cloud providers are investigated and the solutions from each are compared alongside SuperThreads Sheff's on-premises Active Directory solution to determine which is the most effective and cost-efficient.

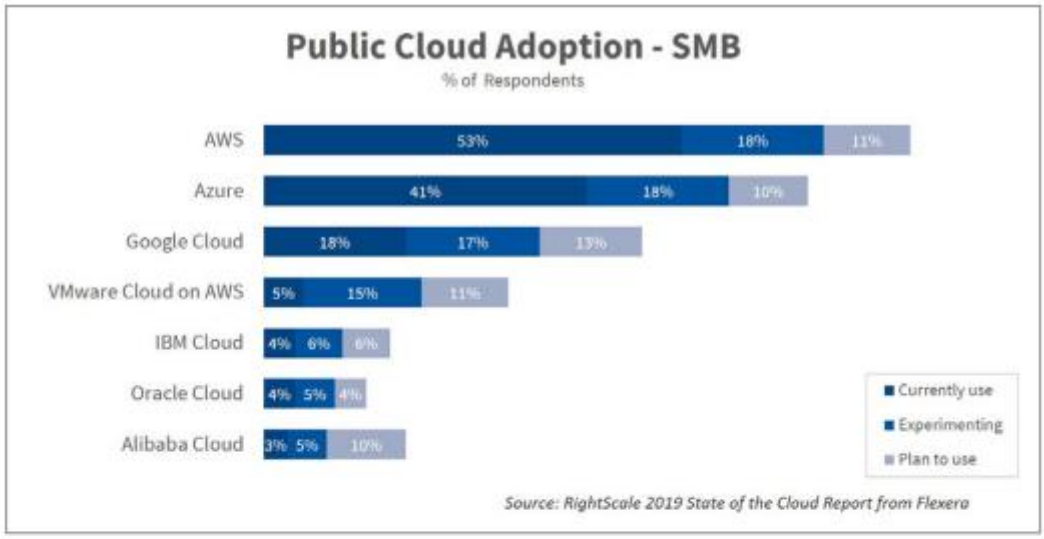
Systems such as Azure Active Directory provide a Cloud-native Active Directory management solution without requiring any of the core on-premise Active Directory infrastructure and is managed completely from the Cloud down to the user (Microsoft Corporation, 2019). Azure Active Directory was first released into General Availability for public use on April 8th, 2013 under the previous name of Windows Azure Active Directory (Guthrie, 2013).

The following three market leading Public Cloud providers are investigated to host the solutions.

- Microsoft: Azure Active Directory Domain Services
- Amazon Web Services: Directory Service
- Google Cloud: Managed Service for Microsoft Active Directory

These three Public Cloud providers were chosen because of the result of a worldwide survey, indicating that more SMB respondents ran applications, were experimenting with services or were planning to use them on these three platforms than other potential public cloud platforms (IBM Cloud, Oracle Cloud and Alibaba Cloud) (Flexera, 2019).

Of the SMB's that responded to the survey, 53% currently used AWS, 41% used Azure and 18% used Google Cloud. 18% were experimenting with AWS, the same percent with Azure and 17% were experimenting with Google Cloud. Of those SMB's not currently using them, 11% planned to use AWS, 10% planned to use Azure and 13% planned to use Google Cloud. As shown in the survey charts on the next page, these three Public Clouds are used and considered more than any other Public Cloud at the time of the report, therefore making them the three most suitable for this investigation (Flexera, 2019).



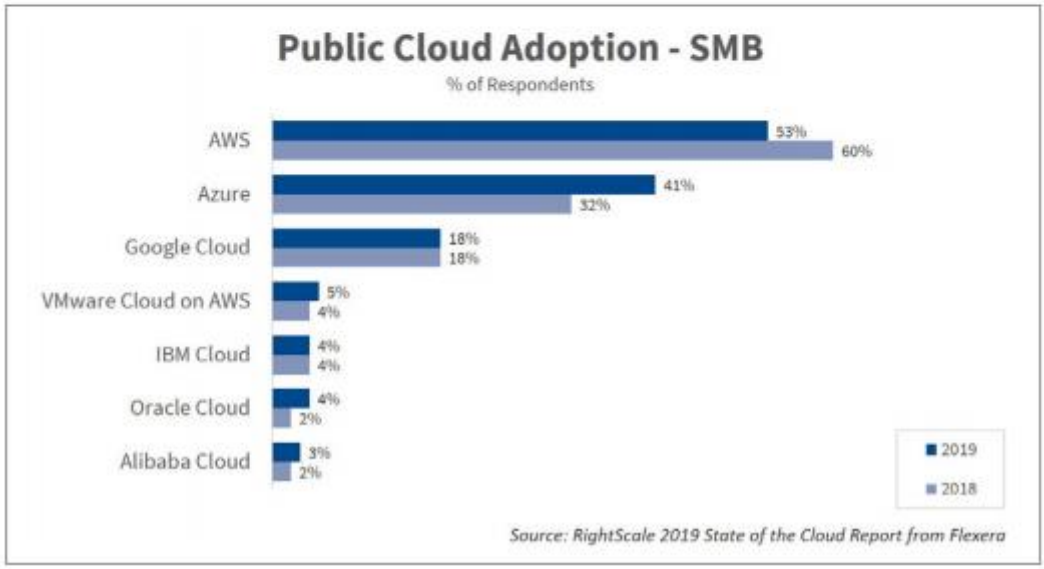


Figure 1 - Public Cloud Adoption statistics for SMB's by RightScale.

The report will show the result of the most suitable Public Cloud provider for an SMB based on the above factors – This report investigates the reasoning behind the popularity behind these Public Cloud providers and what features and cost benefits they offer to an SMB through both a practical and reflective investigation.

Each of the three Cloud platforms has their own offering in terms of an Active Directory management solution. This research investigates each of these solutions, comparing the costs and features between them (Gillis & Rouse, 2018).

Chapter 2 - Literature Review

This section presents and describes the literature that is reviewed and built upon to help conclude the research question at hand. Various topics such as the concept of Active Directory management solutions on-premise through to Cloud based Active Directories and Cloud risks are reviewed here.

In order to fully articulate the project's purpose, the concepts that are aligned with what the project aims to achieve need to be researched.

This begins with the on-premises Active Directory solution to understand what features are in place in a traditional on-premises Active Directory management setting.

The next section of this research looks directly into the Cloud-based Active Directory management solutions that will need to be understood in order to do an investigative comparison between on-premises and Cloud-based Active Directory management solutions.

Throughout this literature review, recent trends and figures around the adoption of Cloud services in general amongst SMB's are investigated. This helps form the basis for research into the more specific topic of Active Directory solutions in the Cloud, rather than a more general approach such as general Cloud cost savings.

On-premises Active Directory management solution

The first question to ask in this investigation is 'what solutions do SMBs' traditionally use on-premises for directory management?'.

To answer this important question, we must take a brief look back into the history of computing with directory services. This historical journey starts with a small document called 'X.500 "Data Networks and Open System Communications – Directory." In which there are a group of standards bodies and vendors from around the world (Boswell, 2003).

The X.500 was a directory service that acted as a distributed store of information about the users of a computer system and the infrastructure that supported that system. Its' goal was to define a single place where users from all nations could locate each other, learn information about one another and communicate freely (Boswell, 2003).

Eventually in the early 1990's, members of the University of Michigan wanted to build a directory service that could handle their above 100,000 students, staff and faculty. This led to the invention of the Lightweight Directory Access Protocol (LDAP). Subsequently, Microsoft chose to adopt LDAP to deliver their directory service, Active Directory (Boswell, 2003).

Active Directory is a feature in Windows Server deployments that allows for the creation of objects that serve as identities and attributes which can be modified dynamically allowing for access, modification or denial to resources and is one of the market-leading LDAP solutions for organisations to manage their identities on-premises (Microsoft Corporation, 2017).

Cloud-based Directory as a Service (DaaS)

In contrast to an on-premises Active Directory solution, the question at hand here is 'how are identities from a directory managed in the Cloud?'.

Microsoft have now developed a Cloud-native way of providing an Active Directory management solution to organisations. Azure Active Directory is their Cloud-based directory management solution that can be purchased and used by organisations (Microsoft Corporation, 2019).

Likewise, Google also has its' own Cloud-based Active Directory management solution called Managed Microsoft AD (Active Directory). Like Azure, it allows an organisation to completely create or migrate their Active Directory solution using Google Cloud (Google, 2019).

AWS Directory Services is a solution provided by Amazon Web Services that allows for a managed Microsoft Active Directory in the AWS Cloud in a similar fashion to the previous two solutions, allowing for a managed Active Directory service ran within the Cloud (Amazon Web Services, 2019).

Risks of operating a Cloud-based Active Directory solution

When dealing with any Cloud service provider, the possible additional risk must be considered as opposed to self-managing hardware and directories on-premise.

The first such risk is the issue of relying on a Cloud provider itself – 'what if the Cloud provider is attacked or has unscheduled downtime?'. Microsoft released its' own statement on the 15th of July 2019 regarding its' Azure Cloud offerings' reliability. It states, "Over the past 12 months, Azure has operated core compute services at 99.9995 percent average uptime across our global cloud infrastructure". It then goes on to reveal "We experienced three unique and significant incidents that impacted customers during this time period, a datacentre outage in the South Central US region in September 2018, Azure Active Directory (Azure AD) Multi-Factor Authentication (MFA) challenges in November 2018 and DNS maintenance issues in May 2019" (Russinovich, 2019). In this instance, the operability of Cloud directory services would be out of the SMB's hands and cause potential loss of business due to being unable to access required services.

There also exists the possibility of the interception of data in transit by malicious entities. Since Cloud resources are off the local premises, any user of these resources within the organisation must form a connection to the Cloud via the internet. Organisations in the United Kingdom are required to comply with the General Data Protection Regulation (GDPR) and are therefore accountable for all data that is stored whether on-premises or in the Cloud and this must be taken into account when trusting a Cloud provider with sensitive information.

Determinants of Cloud Computing adoption

An important question to ask around the topic of this investigation is 'what drives SMB's to want to adopt Cloud-based services in the first place?'.

Recent research into the determinants of cloud computing suggest that there are several benefits including but not limited to; A pay-as-you-go model allowing SMB's to budget better, the elimination of CapEx, faster deployment speeds, scalability and a reduction in cost.

From a pay-as-you-go perspective, the research details that is an important factor for the adoption of cloud services. An SMB can better concentrate on its' actual business rather than worrying about infrastructure requirements, as infrastructure can be scaled vertically, horizontally and to accommodate extra or less computing power and space (such as CPU, RAM, etc.) as and when needed.

Faster deployment speeds mean an SMB can get to market quicker which is a very attractive prospect for SMB's, especially ones that create and host products in the Cloud. With a Cloud-based production system in place, development companies can create and host an application with a few clicks and remove the need for a full development lifecycle to get their application up and running (Parthasarathy & Kumar, 2016).

Cloud Reliability and Agility

Before an organisation of any scale creates or migrates their workloads to the Cloud, an important question to ask is 'how reliable is the Cloud?'. When utilising on-premises resources and infrastructure, it is within the hands of the organisation itself to ensure that their physical and digital assets remain secure and available.

A global network of cloud regions allows for a lower latency for accessing resources and edge locations bring content even closer through deliver networks (Izrailevsky & Bell, 2018). Lower latency access means that when accessing resources, the speed of doing so is much quicker and the upstream and downstream time (upload, download) of resources is also quicker. This would benefit an SMB because they are able to access their resources both on-demand and at great speed.

Chapter 3 - Methodology

This section describes the research methods that are to be used in the investigation as well as what conclusions are arrived at upon completing research into the topic.

It begins with a hypothesis of what the investigation aims to prove followed by the types of research to be undertaken in order to reach a conclusion.

Hypothesis

The hypothesis of the investigation aims to provide an estimated outcome of the research paper, illustrating what the research aims to achieve and how the various aspects of the research will help in answering the questions provided in the aims and objectives section of the introduction.

Creating or migrating Active Directory management solutions in the Cloud will, in theory, save an organisation resources in the form of money and extra space in facilities (due to on-premises server removal or the lack of the need to buy one, for example). It will also provide more features due to its' Cloud-based nature and rolling updates from a vendor as well as taking advantage of the scalability and elasticity that the cloud provides.

The purpose of this research paper is to illustrate the benefits of a Cloud-based Active Directory management solution as opposed to an on-premises solution as well as comparing three of the most popular solutions from Public Cloud providers. As Cloud Computing is a relatively new field, research specifically into the benefits of Cloud Active Directory management services is far and few between and this topic of features and benefits between vendors has seldom been investigated. This paper seeks to resolve that and provide a new foundation for organisations looking to benefit from Cloud-based Active Directory solutions and specifically Active Directory services in the cloud.

Types of Research

There are various research methods that are utilised during this research. Secondary quantitative research will be used to formulate accurate figures, such as analysing the cost of utilising a cloud service from a cloud provider using utilities such as the Microsoft Azure Pricing Calculator which allows an organisation to calculate the pricing of a service running in the Cloud per hour, per day or per month.

Primary quantitative (correlational) research will also be used in the form of gathering prices and figures from energy providers and hardware providers around the local area of the organisation in question, SuperThreads Sheff and calculating them into accurate results using a custom-built tool.

Qualitative research will be used to describe aspects of this research such as when looking into what extra features the Cloud-based Active Directory management solution enables over the on-premises solution and the comparisons between the three Cloud-based Active Directory management solution providers.

Chosen Research Design

The core method of research as a result uses mixed methods to fully illustrate the benefits of Cloud-based Active Directory management solutions from both a qualitative and quantitative perspective.

This illustration is necessary because as previously mentioned, research in the area of Cloud-based Active Directory management solutions is limited. Therefore, Cloud-based directory services are investigated from a new perspective and require Primary Research to be carried out into cost efficiency and feature benefits through the clothing SMB SuperThreads Sheff and using custom made tools where appropriate, that also show the mathematics used in calculating costs which show accuracy.

Showing the correct formulas, code snippets and mathematic equations for this investigation is a key part of inspiring trust in the reader of the research paper (who may also potentially be SMB's looking to benefit from the research from a business perspective) and making replicating the steps in the research and easy task. All the formulas and code snippets are readily available, with the program itself being available on the provided GitHub repository which can be cloned and replicated at any time for use by anyone without a recreational license. This software was created solely for this investigation but can be used or modified in any way required.

Chapter 4 - Formatting the investigation

This section covers the overall format of the investigation, the necessary tools required to research the topic at hand and each stage of the investigation in order or how the investigation will flow.

Designing the investigation

Before conducting primary research, it is necessary to design the flow of the investigation.

Firstly, a proof of concept organisation is used to serve as a baseline in the investigation. This means setting up the organisation with an on-premises server and resources and then carrying out an investigation from the beginning until the end, measuring the cost and features for each Public Cloud provider and then comparing the results to evaluate the benefit and cost differences between them.

A custom-built tool is used to work out the running costs of an on-premises server Windows Server deployment and look at the features of the Windows Server 2016 Essentials edition to compare with Cloud-equivalents on each provider platform.

Creating a baseline for required features and users on the Cloud Active Directory management solution

An Active Directory solution whether hosted on-premises or in the Cloud generally has several identities hosted within it to represent company employees. Because SuperThreads Sheff is an SMB, ten identity objects in Active Directory are used as the baseline for the investigation.

The SuperThreads Sheff organisational diagram shows the ten employees of the SMB in question, their names, and their roles within the organisation in hierarchical order. This is used throughout the investigation when building Cloud-based Active Directory solutions on the three Cloud platforms from a comparative point of view.

It is also important to mention that when looking into creating Active Directory services on each platform, directories that are managed by the Cloud provider (managed directory services) are chosen. This is because on-premises and in the Azure Active Directory services, this is the general baseline for a deployment. On-premises you can choose to use the recommended security features. In Azure AD, these are implemented by the vendor as a native service. Therefore, the most comparable are the managed Active Directory services by each Cloud provider.

Figure 2 - SuperThreads Sheff's organisational chart showing all individuals in the organisation in a ranked hierarchy.

Evaluating the cost of an on-premises server deployment

The starting point of the investigation is evaluating what the cost of running a physical server deployment of Windows Server 2016 Essentials edition is.

This part of the investigation looks at the cost of the barebones server through to the software and features running on it. A custom purpose-built tool for the calculation of physical server costs is used.

Before building this tool, the mathematic equations are drawn out which are required to find the answer to the question ‘What are the costs of running an on-premises Windows Server 2016 Essentials edition deployment?’.

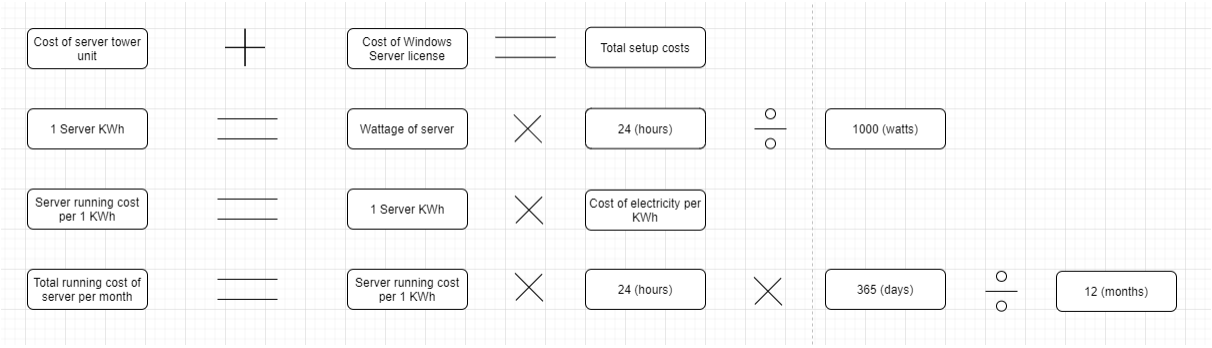


Figure 3 - The mathematical equations used to determine costs and electricity values used in the investigation.

Creating a baseline for required infrastructure on a Cloud platform

In order to make this investigation as accurate and fair as possible it is required that the specifications of any virtual machines ran in the Cloud is identical to the specifications of the purchased on-premises server.

- 8GB of RAM
- 2TB of HDD storage space in Managed Storage
- 1 vCPU
- Any resources must be hosted in a UK datacentre

Additionally, any services must be used on a Pay-as-you-go basis, meaning that no long-term binding contract is signed with the Cloud provider as this can reduce costs artificially. This investigation attempts to use the most accurate figures possible.

Evaluating the cost of services on Microsoft Azure

This stage investigates running and maintaining a Cloud-based Active Directory management service on the Microsoft Azure platform on a month-by-month basis, in terms of costs and the features available on the platform.

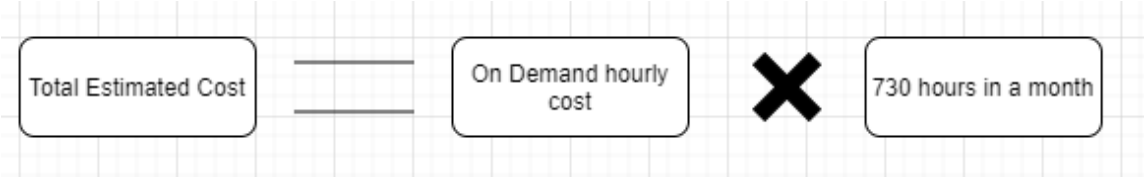


Figure 4 - The mathematical equation used to determine the total estimated cost of running an Active Directory server on Microsoft Azure.

Evaluating the cost of services on Amazon Web Services

This stage investigates running and maintaining a Cloud-based Active Directory management service on the Amazon Web Services platform on a month-by-month basis, in terms of costs and the features available on the platform.

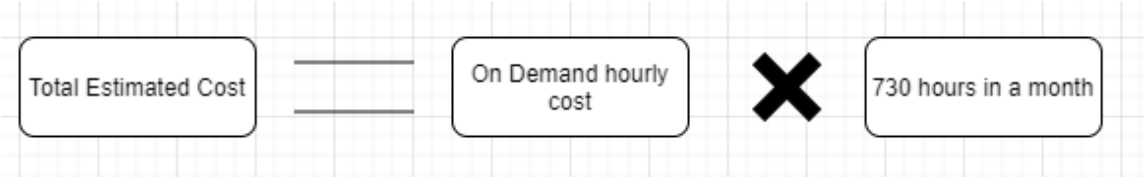


Figure 5 - The mathematical equation used to determine the total estimated cost of running an Active Directory server on Amazon Web Services.

Evaluating the cost of services on Google Cloud Platform

This stage investigates running and maintaining a Cloud-based Active Directory management service on the Google Cloud platform on a month-by-month basis, in terms of costs and the features available on the platform.

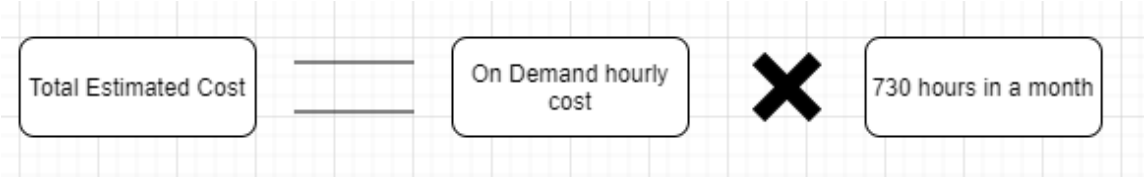


Figure 6 - The mathematical equation used to determine the total estimated cost of running an Active Directory server on Google Cloud Platform.

Evaluating the feature benefits of a Managed Microsoft Active Directory deployment compared to on-premises

The final stage of the investigation design looks at evaluating features benefits of a Cloud-based deployment for SMB's.

Vendors are highly competitive and claim to offer a variety of feature benefits from hosting your solution managed by their Cloud.

Prerequisites and resources used

Yahoo Mail company email address

For the investigation, a company email address is created belonging to the conceptual IT Directory at SuperThreads Sheff using Yahoo's email service. This service is chosen as it is not related to any of the software vendors being investigated and therefore would only serve to create on-premises identities during the architecting of the on-premises directory solution.

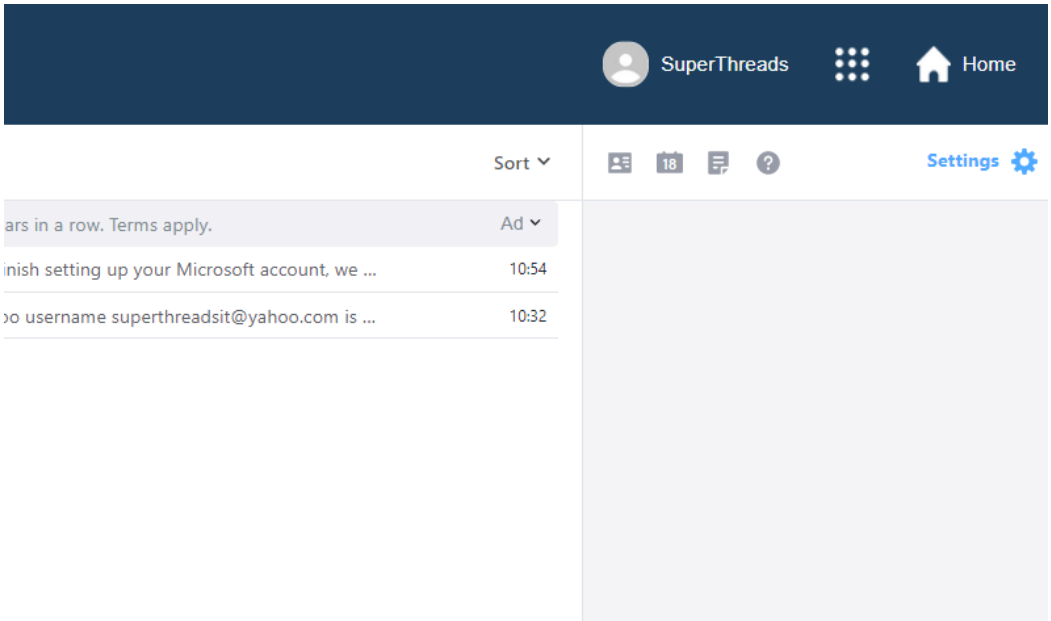


Figure 7 - The Yahoo mail account set up for SuperThreads Sheff as part of the investigation.

Windows Server 2016 Essentials edition license

The core component of an on-premises SMB server is the Windows Server 2016 Essentials edition license, which allows for the creation an Active Directory deployment of organisational users and the ability to administrate them on-premises, allocating organisational users with a digital identity.

As the experiment does not take long, the Microsoft 120-day evaluation period was used, and this allowed for the utilisation of the full extent of the software up until that point in time. This is enough time to conduct the experiment using the software. The reader of the investigation can at any time visit the downloads page of the Microsoft website and download an evaluation version of the software used in this investigation to replicate the investigation or for their own purposes.

The reason the Essentials edition of Windows Server 2016 is chosen is because of its' description by the vendor, Microsoft, on their official technical documentation blog which states that it is suitable 'for small businesses with up to 25 users and 50 devices (Microsoft Corporation, 2016). Additionally, a separate industry blog describes the various versions available on the market and states that 'picking the right version can be the difference between saving a lot of money and losing a lot of money', which is first evidenced by the actual upfront cost differences in editions (NetworkProGuide, 2018). It is therefore deemed the most cost-efficient version of the software and the most logical to use for an SMB.

Microsoft Azure Pricing Calculator

In order to best establish a set of figures for comparison with the other potential solutions the official Microsoft Azure Pricing Calculator is used. This calculator is used by organisations that are looking to either migrate or create their solutions in the Cloud and is already an established service offered by the organisation free of charge (Microsoft Corporation, 2020).

Amazon Web Services Pricing Calculator

Amazon Web Services also has an established pricing calculator for organisations who are considering migrating or creating services in its' Cloud. This will be used in the investigation to calculate the costs of creating and maintaining an Active Directory solution in the Cloud (Amazon Web Services, 2020).

Google Cloud Platform Pricing Calculator

Google Cloud Platform, much like the previous two Clouds, has a pricing calculator that serves the same function. This will be used to investigate the costs of creating and maintaining an Active Directory solution in the Cloud (Google, 2020).

Artefacts

ServaCalc, a custom-built Python tool for calculating the physical running costs of a server on a month-by-month basis

ServaCalc is a custom self-developed Python tool that is used to calculate the costs of running a server of a certain specification over the course of a month (Hawkins, 2020). The link to this tool can be found in the references appended at the bottom of this investigation.

The tool is dynamic in that when run, it can take the input of the information from any server and give a reading for the cost of running this server per month. This gives me primary research information to use as I can compare the results of this with Cloud-based costs. For the purpose of this investigation, the tool is used to evaluate the costs of a server targeted at the SMB audience running a Microsoft Windows Server 2016 Essentials edition license.

```
try:
    print("Welcome to ServaCalc, the simple server running costs calculation tool. I need to take a few bits of
    towerCost = float(input("What is the cost of your tower unit in pounds and pence?\n"))
    wattValue = int(input("How many watts does your tower unit use? See the manufacturer's website for details.
    electricityCost = float(input("Please enter the cost of your electricity in the form of pence per KWh. Chec


except ValueError:
    print("Please only enter digits.")


else:

    kwhHour = wattValue * 24 / 1000
    kwhHourCost = float(kwhHour * electricityCost)
    totalcostMonth = float(kwhHourCost * 24 * 365 / 12)

    print(f"Your initial setup costs are £{towerCost:,.2f}".replace('$-', '-$'))
    print(f"Your total monthly server running costs are £{totalcostMonth/100:,.2f}".replace('$-', '-$'))
```

Figure 8 - The code used to build the ServaCalc program in Python for use in the investigation.

This software was built entirely from scratch using Visual Studio Code, which is a free code editor built on open-source software. This code editor can be downloaded at any time and used to replicate the above code, whether to use the code or to make changes to the code itself. Visual Studio Code can be downloaded from the official Visual Studio website (https://code.visualstudio.com) (Microsoft, n.d.).

Chapter 5 - Implementation and Results

This section is used to display the implementation of the work described in previous sections and to show the results of the investigation.

In this section, the overall cost of the on-premise and each Cloud solution is shown as well as the steps taken to get there, and the research question is answered through analysis of the results.

Choosing an on-premises deployment, architecting the solution and evaluating initial costs and costs to run per month

The first result required in order to compare costs between on-premises Active Directory solutions and those running in the Cloud is the cost of running a server on-premises.

So, the first step in finding this result was to find an appropriate server that would be suitable for an SMB and load the identity solution onto it. Research was done using online articles that looks into the cost and types of servers available in the UK market and this showed that Lenovo offered various standing server towers aimed at SMB's (Gildred, 2018).

In choosing a server to use for the experiment, one was chosen that was as up to date as possible that a new SMB would consider purchasing to host their identity solution. This server was the ThinkSystem ST50, which is described as 'a powerful, yet whisper quiet, entry tower server ideal for retail, remote/branch or home office… Featuring enterprise-class storage, reliability and serviceability support, the ThinkSystem ST50 is the ideal first server for the growing SMB or distributed work environment' (Lenovo, n.d.).

Finding the cheapest possible price for a Windows Server 2016 Essentials license

A cost that must be considered alongside the physical hardware in this investigation is the licensing itself. For SuperThreads Sheff to run an identity solution on their server, they are going to require a license to do so.

In this case, the cheapest possible single license price for Windows Server 2016 Essentials was found on Lizengo's website and was used as a value that was entered into the ServaCalc program alongside the tower cost. The price of the Windows Server 2016 Essentials license was £239.99 (Lizengo, n.d.).

Figure 9 - Lizengo webpage showing the cost of a Windows Server 2016 Essentials license for use in the investigation.

Finding the tower cost (initial setup cost) and the wattage output

In terms of finding the cost for the tower itself, a single unit for sale had to be found, as Lenovo mainly deals with enterprise customers individually and does not display a set price. This was found on serversdirect.co.uk, a UK retailer for servers and the price was taken from this website as a figure to put into the initial costs field in the ServaCalc tool – the cost being £679.97 (inc. VAT) (serversdirect, n.d.). The values used in the tool and for comparison can literally be any server of any specification, however through research it was found that this server is one that fits the needs of SuperThreads Sheff as an SMB.

In order to find pricing per KWh, the wattage output of the tower needs to be used, so through the technical specifications of the server on the Lenovo website it is shown that it runs at 250 watts with maximum power efficiency (Lenovo, n.d.).

Power	1x fixed 80 PLUS power supply up to Platinum efficiency; 250W or 400W

Figure 10 - The power figure in Watts for the server used in the investigation from serversdirect.

Finding a figure to use as a KWh value for the investigation

In order to find a figure to use as a KWh value for the ServaCalc program, an energy provider is chosen at random in the Sheffield postcode area (British Gas) and checked for a quote on business energy.

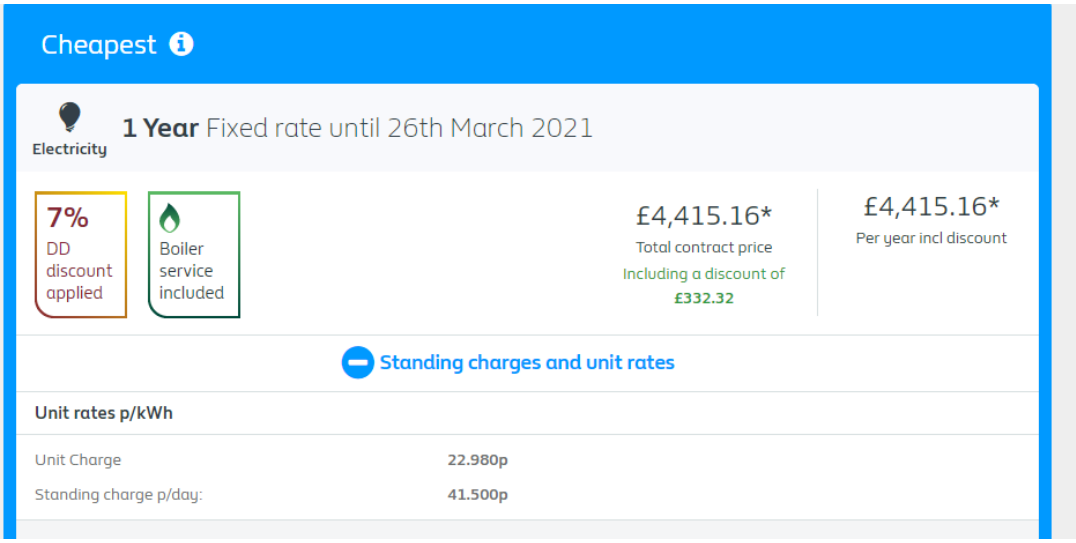


Figure 11 - Electricity prices for the postcode in which SuperThreads Sheff is based (Sheffield, UK).

As depicted in the quote, the unit charge per KWh (p/KWh) is 22.98p (British Gas, n.d.). Therefore, this is used as the KWh pence value in the ServaCalc program.

Utilising the ServaCalc program to calculate the total on-premises server costs per month

The next stage of this part of the investigation involves loading up the custom-built tool program and giving it the figures that were researched as above to produce an accurate reading of the cost per month for running the on-premises server in question, which in theory would run the Windows Server 2016 Essentials edition and act as the sole identity server for SuperThreads Sheff.

Using the Ubuntu subsystem for Windows 10, the ServaCalc python program is run from the command line and is given these figures.

```
curtis@CurtisWorkstation:~$ python3 servacalc.py
Welcome to ServaCalc, the simple server running costs calculation tool. I need to take a few bits of information from yo
u.

What is the cost of your tower unit in pounds and pence?
679.97
What is the cost of your Windows Server license?
239.99
How many watts does your tower unit use? See the manufacturer's website for details.
250
Please enter the cost of your electricity in the form of pence per KWh. Check your energy provider for details.
22.98
Your initial setup costs are £919.96
Your total monthly server running costs are £1,006.52
curtis@CurtisWorkstation:~$
```

Figure 12 - The ServaCalc python program running from the command line, displaying costs for the investigation.

The result of the calculations produces the following results:

- Initial setup costs: £919.96
- Total monthly running costs: £1,006.52

A spot rate conversion to USD from GBP using Refinitiv's market data allows me to convert this value for the purposes of comparison (Refinitiv, 2020).

- Initial setup costs: $1,130.72
- Total monthly running costs: $1,237.11

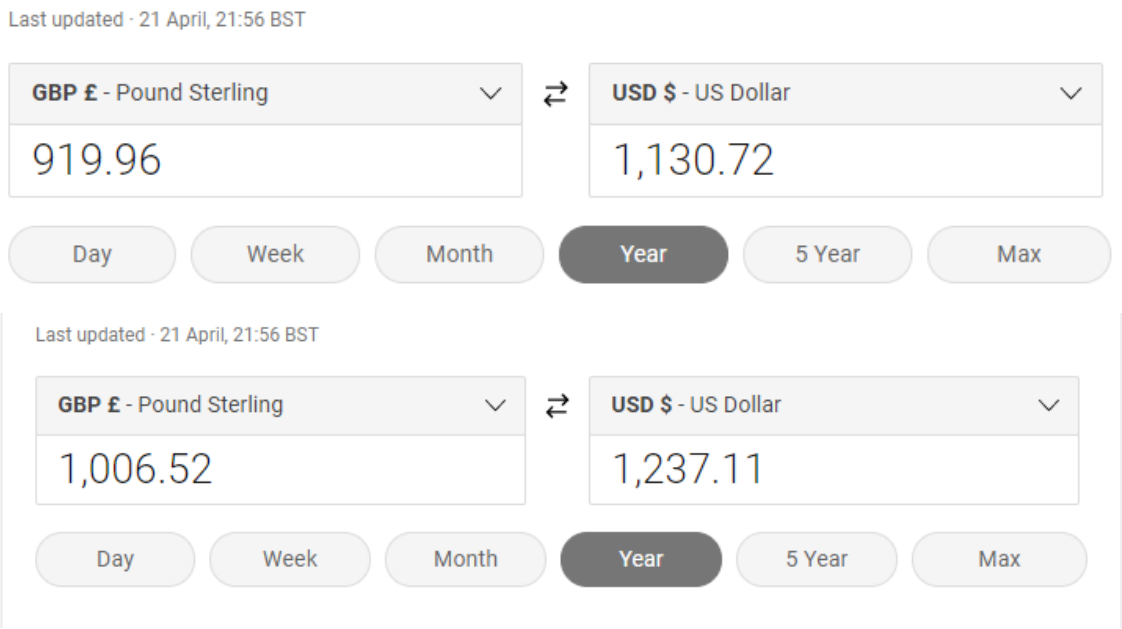


Figure 13 - Spot rate conversion between GBP and USD.

Installing the Windows Server 2016 Essentials edition onto the server

A major part of preparing for the investigation was installing the correct version of Windows Server 2016 (Essentials) and then creating the identities within the program running on a server. The server specifications as shown above are used and only the necessary users and roles are installed.

The exact steps of this installation can be found at the end of this paper in the setup manual for the installation of Windows Server 2016.

Calculating the costs of running the equivalent solution on Microsoft Azure

Using the benchmarks outlined previously in the investigation, the configuration figures are entered into the Microsoft Azure Pricing Calculator in order to provide an estimated cost.

The initial instance figures are entered to match the specifications laid out in Section 5.5.4 around the technical specifications of the hardware and the pricing is set to a Pay-as-you-go model.

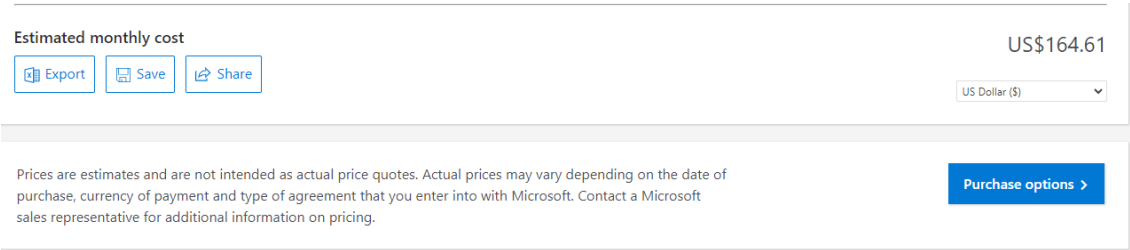


Figure 14 - Screenshots of cost calculations using the Azure cost calculator.

The final price comes out at $164.61.

Calculating the costs of running the equivalent solution on Amazon Web Services

Using the benchmarks outlined previously in the investigation, the configuration figures are entered into the AWS Pricing Calculator in order to provide an estimated cost.

As with the previous Cloud provider, a quick estimate is used to figure out the general cost of quickly provisioning and setting up an Active Directory server with 2TB of Cold Storage, the most cost-efficient HDD storage available in this tier.

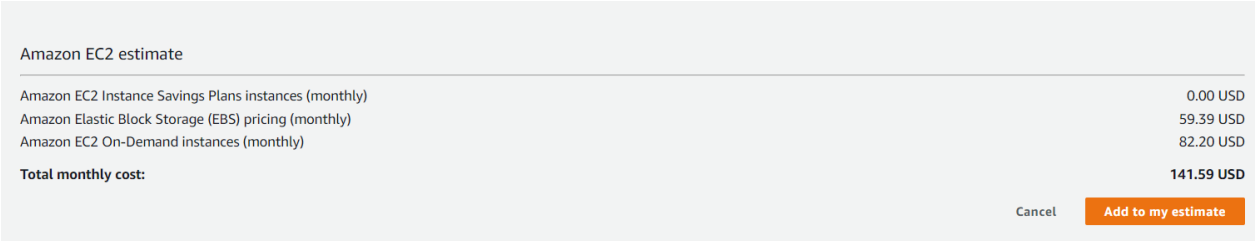


Figure 15 - Screenshots of costs using the Amazon Web Services cost calculator.

As depicted, the most optimal instance offered aligned with the specifications laid out in Section 5.5.4 is used. The On-Demand pricing strategy is used to declare that SuperThreads Sheff does not intend to sign into a binding long-term agreement but would rather pay as they use the services. The figures are then calculated using the calculator.

It is found that the total monthly costs for using an AWS virtual machine instance to host Active Directory is $141.59.

Calculating the costs of running the equivalent solution on Google Cloud Platform

Using the benchmarks for specifications stated in Section 5.5.4, these benchmarks are entered into the Google Cloud Platform calculator to estimate the costs of running the solution.

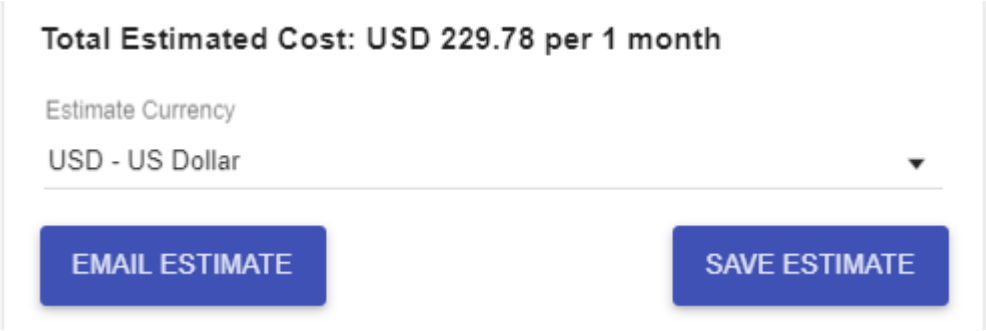


Figure 16 - Screenshots of cost using the Google Cloud Platform cost calculator.

As depicted, using identical specifications the total monthly cost comes to $229.78.

Comparing the monthly costs between the on-premises deployment and each Cloud provider

As all the figures have been gathered from each stage of the investigation, the cost comparison for each solution's running cost is tallied below.

Upfront Costs:

- On-premises - $1,130.72 upfront.
- Microsoft Azure – No costs incurred.
- Amazon Web Services – No costs incurred.
- Google Cloud Platform – No costs incurred.

Monthly Costs:

- On-premises - $1,237.11 per month.
- Microsoft Azure - $164.61 per month.
- Amazon Web Services - $141.59 per month.
- Google Cloud Platform - $229.78 per month.

In terms of overall costs, the Amazon Web Services platform is determined to be the most cost-effective method of hosting a Windows Server Active Directory solution.

This investigation shows that not only is one Cloud provider more cost-effective than its' market leading competitors, it is also much more cost-effective than an on-premises solution.

Comparing the feature benefits between Active Directory managed in the Cloud and on-premises

Each Public Cloud provider offers various benefits when utilising what is known as a Managed Microsoft Active Directory instance. The verb managed refers to the fact that the Cloud provider manages most aspects of this deployment for an organisation utilising the platform.

A Microsoft MVP (Most Valuable Professional) describes this process on Amazon Web Services, stating 'AWS Directory Service allows you to run Microsoft Active Directory (AD) as a managed service within the AWS Cloud. The underling Active Directory infrastructure is 100% managed by Amazon Web Services, equating to an extremely simple deployment and management process (Wyckoff, 2018).

Azure Active Directory Domain Services

Since Active Directory is a Microsoft product, managed service features are automatically provided when deploying a Microsoft OS (not just Windows Server) into Microsoft Azure. In relevance to an SMB, Microsoft claims the following features and benefits of using Azure Active Directory Domain Services in this manner:

- Simplified deployment
- Integration with Azure AD
- Usage of corporate credentials
- High availability

A single wizard is required for the enablement of a tenant, making the process of signing up for the service quick and simple. Integration with its' Cloud-native Azure AD platform may make things easier in future for an SMB who wishes to utilise this platform. Since corporate credentials can be used in the Cloud, this makes it easier for an organisation to manage. High availability means that the SMB will have a more redundant solution in the Cloud, as their Azure Active Directory Domain Services deployment includes multiple domain controllers and guarantees service uptime and resilience to failures (Microsoft Corporation, 2020).

AWS Directory Services

Amazon Web Services' offering claims the following features and benefits for their managed instances of Microsoft Active Directory. In relevance to an SMB, the benefits are:

- Easily migrate directory-aware, on-premises workloads
- Use actual Microsoft Active Directory
- Share a single directory for cloud workloads
- Centrally manage application access and devices in the AWS Cloud
- Simplify administration with a managed service

If an SMB chooses to migrate their Windows Server deployment, this is easily done on the platform. All tools that were previously used would be the same as in the AWS Cloud upon migration. A single directory structure makes it simpler to manage Cloud workloads and multiple IT management accounts can be used for one directory, which also helps to avoid complexity in replicated and synchronising data across multiple directories. Application access and device management allows an SMB to use existing device credentials amongst staff and access cloud-based resources while remaining mobile. Being a managed service, the SMB gets access to a deployment across multiple Availability Zones, providing better availability for failure while AWS also manages all patches and software updates (Amazon Web Services, 2020).

Google Cloud Platform Managed Service for Microsoft Active Directory

Google Cloud Platform's offering claims the following features and benefits for their managed service of Microsoft Active Directory. In relevance to an SMB, the benefits are:

- Compatibility with AD-dependant applications
- Virtually maintenance-free
- Seamless multi-region deployment
- Hybrid identity support

As the service runs real Active Directory Domain Controllers, compatibility for applications is not an issue and standard Active Directory features and management tools can be used like on-premises. The service is highly available, automatically patched and configured with secure defaults whilst being protected by appropriate network firewall rules. The service can be deployed in a specific region to allow apps or virtual machines in the same other regions to access the domain over a low-latency Virtual Private Cloud (VPC) and this allows an SMB to grow their infrastructure as needed by expanding to additional regions. An on-premises Active Directory domain can be connected to the Google Cloud or deployed as a standalone domain for cloud-based workloads (Google Cloud, 2020).

Benefits of a managed Cloud-based solution to on-premises

As detailed in the previous sections, the benefits of using a purely Cloud-based or even a hybrid Cloud Active Directory management solution compliment the cost benefits of doing so very well. Not only is the Cloud considerably less expensive than on-premises for managing an Active Directory service as an SMB, but also much more feature rich.

The aspects of this outlined in the previous sections are scalability and reliability. Each of the Public Cloud providers mentioned includes access to multiple Availability Zones or failover Domain Controllers for an SMB organisation so that if the deployment were to fail for some reason, they are guaranteed an SLA and uptime in continuing their business on another Domain Controller or within another Availability Zone in which their failover and backup resources are hosted.

A managed solution is also easier to maintain for an SMB, as management of the updates, patches and provisioning of extra resources elastically is done by the Cloud Service Provider (CSP) rather than the SMB – which saves them a lot of time and money.

Chapter 6 - Discussion

This section covers the discussion in which the new contribution this report provides to the domain of knowledge is demonstrated and points from the various areas in the report are discussed.

The investigation set out to prove that a Cloud-based Active Directory management solution is more cost-effective than an on-premises deployment for an SMB.

The results that were found during the investigation clearly show a rather large cost saving from deploying and running Active Directory in the Cloud. When Public Cloud providers first started offering services on their platforms, a major selling point for them was that these services would be much more cost-effective compared to an organisation having to purchase the physical infrastructure required to run the solution as well as maintaining the physical servers powered on 24/7 for 7 days a week, 365 days a year. This report sought to investigate that claim pertaining particularly to Active Directory management services in the Cloud.

Technically, the figures produced during this investigation can vary from time to time due to cost changes on behalf of the Cloud provider for their platforms. Also, the calculators that are provided by each Public Cloud provider are clearly stated as rough estimates and can be negotiated with each Cloud provider. For example, Microsoft Azure claims to price match with Amazon Web Services on comparable services (Microsoft Corporation, 2020). The Public Cloud provider landscape is that of a competitive environment, with providers competing against each other constantly to offer the best prices, with prices fluctuating constantly to reflect this.

Therefore whilst this investigation does not claim to be concretely correct about the figures provided in terms of price, it can be deduced from the investigation that a Cloud-based solution is much more cost-effective than an on-premises solution when hosted on every investigated platform (Microsoft Azure, Amazon Web Services and Google Cloud Platform).

Whilst the original research question was answered and the point of a Cloud-based Active Directory management solution being more cost-effective than an on-premises equivalent was proven in terms of the sheer margin of price difference, there are still various weaknesses and gaps in the research. For example, this research does not go deeper into alternative identity management solutions such as Azure Active Directory, which can act in a similar manner to Windows Server's identity features within Active Directory Domain Services without requiring a physical server present at all – being completely Cloud native (Microsoft Corporation, 2019). Similarly, there are other alternative solutions than Active Directory altogether – such as the G-Suite (Google Suite) offering which can serve to provide identities for organisational users and other functions such as organisational emails and directories (Google, 2020).

The investigation was used to gather figures for Cloud-services on a Pay-as-you-go basis. An organisation can save more money by choosing to commit to a longer-term basis with a Public Cloud provider. This report did not investigate this further, as because previously stated, Public Cloud providers are highly negotiable on prices and often compete for customers to sign up and utilise their platform. This is what could be considered another weakness of the investigation because figures are constantly in motion and subject to change, which technically makes the figures generated by the investigation invalid past the date of original estimation.

Therefore, while it was found that Microsoft Azure offered the most cost-effective Windows Server deployment option during the investigation, this could change at any time between the three providers. The major reason that Microsoft can offer a more attractive price point is also because of their ownership of the product licensing itself. Windows Server after all is a product of Microsoft. But what is to state that the infrastructure of the other two providers does not become cheaper to run for the provider themselves in the future? This could drastically change the results of this investigation. Because of this, the investigation clearly details each step in order to allow references to replicate the investigation in future, through the custom-built artefact tool that allows for the calculation of on-premises server and licensing costs to the setup manual at the end of the report that details setting up identities on an on-premises server. It is also made clear through the report on how to estimate costs on each Cloud platform.

Another aspect to consider would be staff capability. Cloud-based skills are relevantly new and still on the rise, so IT staff of the organisation may not already have the Cloud administration skills required to either build or maintain this solution, or both. This is a factor unique to each SMB and would need to be considered case-by-case, as it may work out still less expensive to actually train staff on the Cloud skills required to implement and maintain this solution than the upfront costs of an on-premises deployment and the monthly running costs.

It is made clear during the research that the feature benefits complementing the cost-efficiency of running an Active Directory management solution in the Cloud are rich and vast. In a traditional on-premises deployment, it is expected that IT staff at the SMB in question are required to maintain patches and security upgrades to physical servers, as well as making sure that there are redundancy measures in place in case of a hardware failure that could be catastrophic to the business – resulting in a loss of service for organisational users that could consequently cause the organisation to lose business if they are unable to access required resources within Active Directory.

Chapter 7 - Conclusion

This section concludes the project. This project sought to build the gap in knowledge between SMB's looking to build their vital core infrastructure with Active Directory and the platforms available for them to do so.

It details cost differences and feature benefits whilst comparing three market leading Public Cloud platforms in terms of service offerings. It is intended for use by an SMB who is looking to use Active Directory and may have been looking at alternatives to on-premises solutions and require guidance on Cloud-based solutions as well as contributing to literature on Cloud-based services as opposed to traditional on-premises services, as per the literature reviewed during the investigation which suggest that organisations around the world are increasingly adopting or looking to adopt Cloud-based services for their organisations as opposed to on-premise.

The research proves that on-premises costs, in terms of the up-front cost required and the maintenance costs of running a server consistently are much more expensive than that of simply building the solution on a Cloud platform, regardless of which is chosen by a large margin.

It was found practically impossible to concretely prove that one Public Cloud provider is better than another in terms of costs from a neutral point of view at the time of the investigation. As previously stated in this investigation, these prices are constantly fluctuating and can mostly be negotiated with each provider, taking additional factors into account such as account longevity with the provider and mutual interests between the SMB and provider. Additionally, it should be taken into consideration that the cost differences for equal infrastructure on each Cloud platform were not massively disperse.

So, it is beneficial for an SMB to use the Cloud when they are building an Active Directory solution from the ground up in terms of both costs and features. An organisation would have to both purchase upfront hardware and maintain infrastructure within their organisation, accounting for any failures and redundancy measures as well as making sure staff are constantly patching for security and general updates should they deploy on-premise. Every Public Cloud provider investigated in this document negates this issue by providing a fully managed service to take these steps for the SMB.

Chapter 8 - Critical Reflection

This section is a reflection on what happened during the investigation as well as what I have personally learned and how I have developed through carrying out this project to completion, from concepts in my degree course through to information learned by experimentation.

Further potential research topics in relation to this investigation

What are some further potential topics that can be researched in relation to the investigation carried out in this report going forward?

Alternative Cloud-based solutions as opposed to on-premise

Windows Server and consequently Active Directory management solutions are not the only services that can be hosted in the Cloud rather than on-premises. Further research can be undertaken into other services, such as for example productivity applications in the Cloud or virtual machines and networks. This investigation was a single solution as opposed to potentially thousands that can be investigated in this way on a service-by-service basis.

These services can range things such as databases, code as functions (serverless code), entire websites and more.

Larger organisations in relation to on-premise versus Cloud-based

This investigation focused purely on the cost-efficiency and benefits of an SMB and used resources and licensing targeted at SMB's. Further research can be done into this topic on a larger scale in relation to larger organisations, with potentially hundreds and thousands of services rather than a single service. The potential scale of Cloud services for larger organisations are huge – with larger organisations often using thousands of Cloud services across multiple Cloud tenancies of various types – such as databases, web services and functions as code.

Acquisition of research skills and the third person point of view

The overall final year project was the very first time I stepped into the world of research. At first, I found it quite difficult to frame the research in the third person and attempt to be as neutral as possible when conducting the research and presenting findings in the body of the document.

Originally, I had written down most of the body of the report in the first person, using verbs in the first person such as 'I put the figures into the calculator' – It took a fair amount of time to review back over the document and learn to frame in the third person, making edits and sometimes even starting sections again where I thought I had too much of an opinion on the topic at hand and instead looked to present factual information.

I have learned to find and reference information from a range of sources, including research papers that can be easily enumerated through Google Scholar or sections of published books.

Re-framing the investigation and starting again due to technical and ethical limitations

When I first started carrying out my research for the investigation, I looked at each of the Cloud providers from a more technical standpoint. I was originally planning to sign up for each of the Cloud providers using a conceptual email address that I created in Yahoo for the organisation. As I was signing up and looking at each of the Cloud platforms, I ran into various issues.

When signing up for each Cloud platform, a user is generally given trial credit and a trial period. As I started building workloads in the Cloud such as VM's utilising Windows Server, I realised that this would not suffice to give me the cost figures I needed on a month-to-month basis. Certain Cloud providers such as Microsoft Azure only allow premium subscriptions or paying users to access their monthly cost predictions, so this became non-viable to do.

Before even starting the project I also originally planned to compare Cloud-native identity solutions such as Azure Active Directory to Active Directory running on Windows Server. This was a learning curve for me, as I came to realise that these were two completely different concepts and not comparable from a feature or cost perspective. This expanded my depth of knowledge into Cloud and on-premises identity and how Windows Server plays a role in providing identity management services to an organisation through Active Directory. This is when I then re-framed my investigation and decided to investigate Active Directory solutions hosted in the Cloud rather than directly serviced from the Cloud (Cloud-native applications).

Better understanding of Cloud Computing

This investigation served as a good practical and theoretical knowledge gaining experience in the domains of Cloud Computing from a commercial and technical aspect.

Organisations are constantly looking for new talent in Cloud Computing to expand their own knowledge and better serve their customers, so this has contributed to my readiness in this domain and could make me more employable in the future.

Cloud Computing is a constantly growing field within information technology and being able to adapt core IT skills into the Cloud is a valuable tool in any IT practitioner's skillset.

Programming and technical problem-solving skills

Throughout my education, I had used programming skills relatively little. This investigation served as a great practical challenge in creating an original piece of code that solved a problem. This code can now be re-used in future investigations and has contributed to both my skills as a programmer and my portfolio.

Creating the ServaCalc tool was the first time I utilised the GitHub platform, which will aide my future development as I aim to create more programs in my study and work which can be showcased on my GitHub portfolio page. By creating my own program for this project, I was introduced to Visual Studio Code, GitHub and version control when developing software (https://github.com/Curtish96) (Hawkins, 2020).

Using what I had learned in my degree to aide me in this project and a reflection of the concepts learned throughout my degree

Sheffield Hallam University, in my opinion, gave me a strong grounding for practical skills and experimentation. The university has various laboratories and physical equipment set up for experimentation so that concepts learned through teaching can be practised.

It was through learning how to effectively experiment with computing concepts that I looked into Cloud Computing originally and developed my understanding of it – this began with what I learned about Active Directory whilst undertaking a second year module in Network Services and Administration. I learned about how user identities work on-premises and how Active Directory acts as the hub for user identities and accounts in an organisation. This led me to ask the question – “How can this be made more effective? How can an SMB best take advantage of this? What are the alternatives?” – which then helped me arrive at my solution of looking at hosting Active Directory in the Cloud.

Another core technology used in my project as previously mentioned was Python. My first look at programming was in a basic software development fundamentals module at college but never went much deeper into actual programming languages or how they work. In my first year of university, I was taught about Python as an object-oriented programming language (OOP) and how it can be used to solve problems. This led to me building a custom tool in the language and using it to work out costs for physical hardware for this project, which is now hosted on my GitHub account should it need to be further reviewed and can be seen or adapted by anyone in the world.

The Computer and Information Security undergraduate degree course has prepared me for further research in the IT industry, whether in the domain of cyber security or more generalist and has allowed me to obtain a valuable core skillset that I can use in industry or further education.

The project itself has given me a grounding for further research into a topic that vastly interests me – Cloud Computing. I can apply this knowledge in industry to further myself and my colleagues in this rapidly evolving discipline. As previously stated in the section referencing ServaCalc, I have also developed a piece of software for the very first time and learned about code editors and version control software – this is something I would also like to continue to learn about and improve my programming and development skills in future.

Chapter 9 - On-premises setup manual (Active Directory)

This additional section walks the reader through the steps required to create their own on-premises deployment of Windows Server 2016 Essentials edition and set up the required features, such as creating user identities and a network administrator account that is used to administrate the deployment.

After purchasing a server and booting it, the next stage is to install and utilise Windows Server 2016 Essentials edition to enable identity features for the organisations' employees.

The server boots with the 'Configure Windows Server Essentials' prompt and asks for company information, in which the details for SuperThreads Sheff are entered.

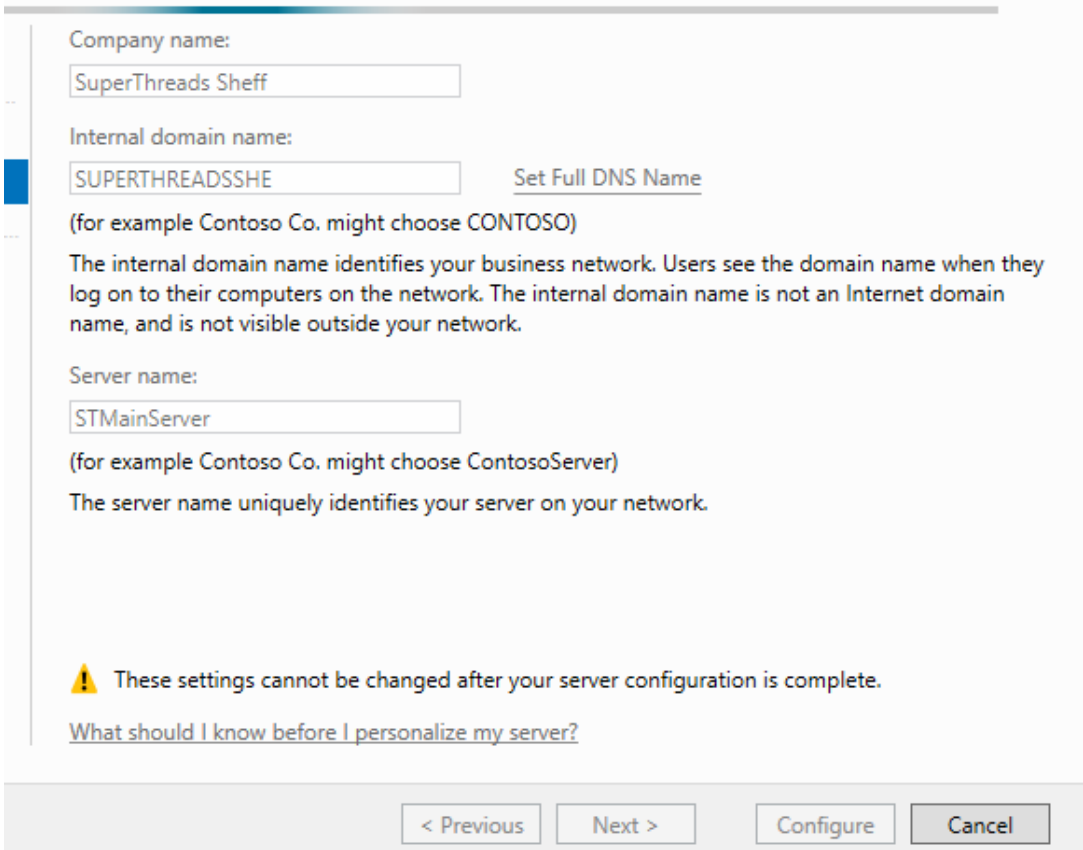


Figure 17 - Setting up Company information in Windows Server Essentials.

The Administrator account name is then entered which is that of the IT Director, Selena Bass who oversees the IT Operations at SuperThreads Sheff.

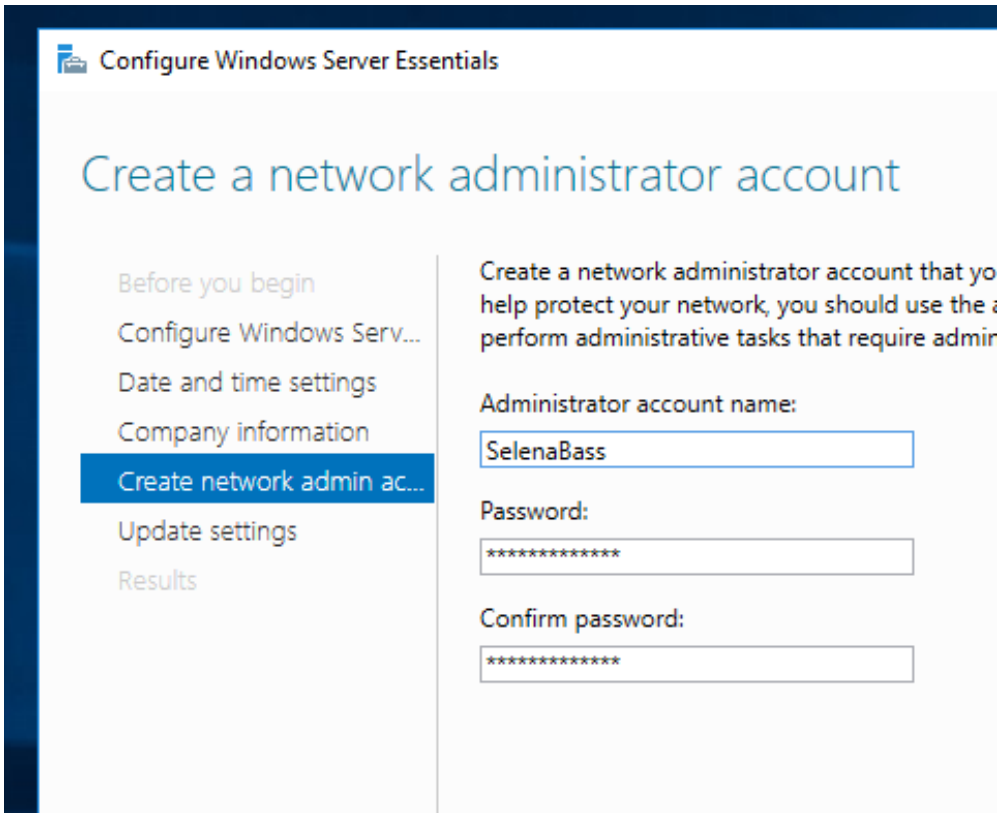


Figure 18 - Creating an administrator account in Active Directory.

The recommended settings from Microsoft are used to minimise any security vulnerabilities from the get-go, advice given at the prompt.

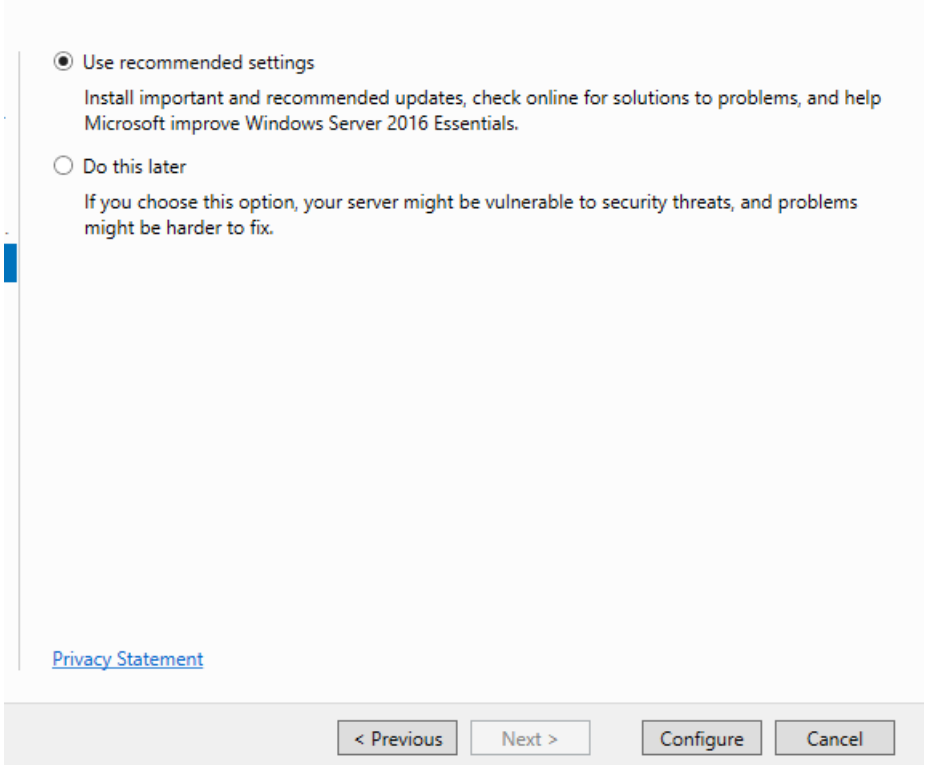


Figure 19 - Selecting settings to use in Windows Server Essentials.

After this, the server finishes its' configuration and architecture of the on-premises solution can begin.

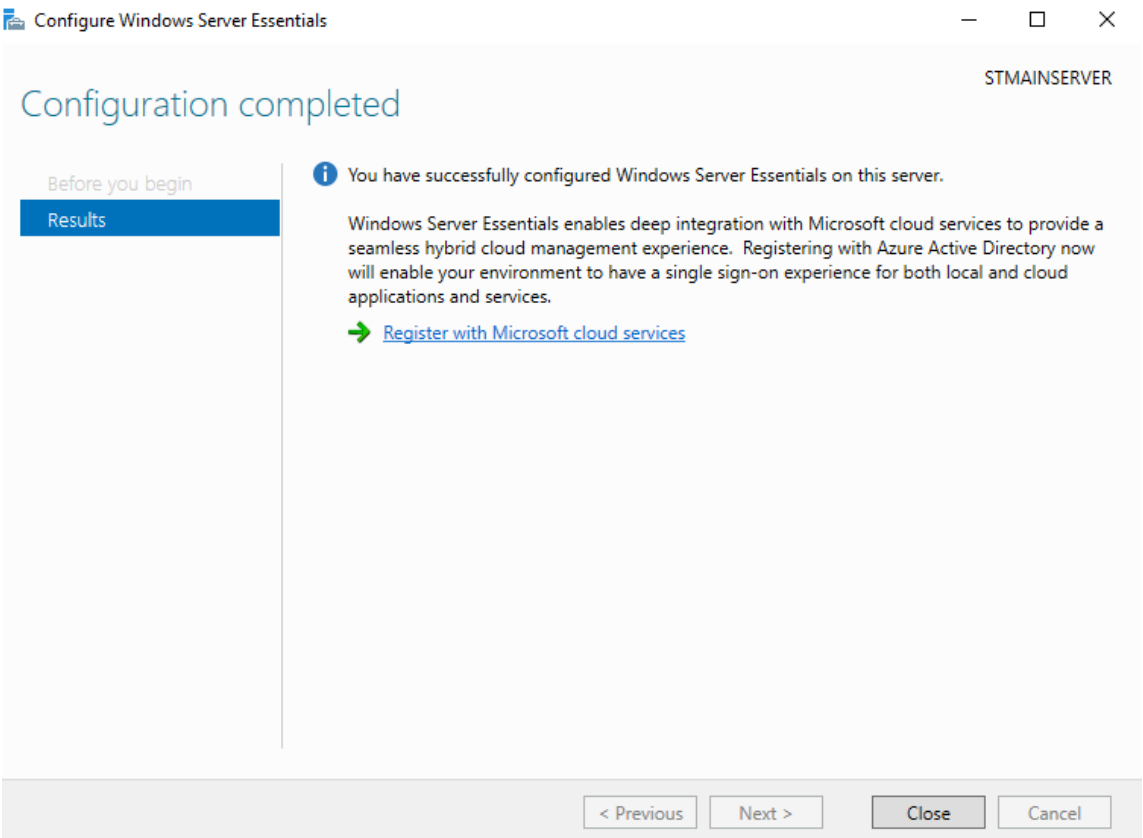


Figure 20 - Windows Server Essentials completing configuration.

The Windows Server Essentials dashboard allows for the setup of any initial variables within the deployment such as user accounts and other connections and features.

The main feature used here to complete setup of the organisational identities is the 'Users' and 'User Groups' creation functionality, which is done via the 'Users' dashboard.

The most straightforward method of creating users is to use the 'Add a user account' functionality under the 'Users' dashboard as it is guided by the GUI. The account required to do this is an Administrator account and for the purposes of this investigation we are using the account of Selena Bass, the IT Director, as would be done in a real scenario.

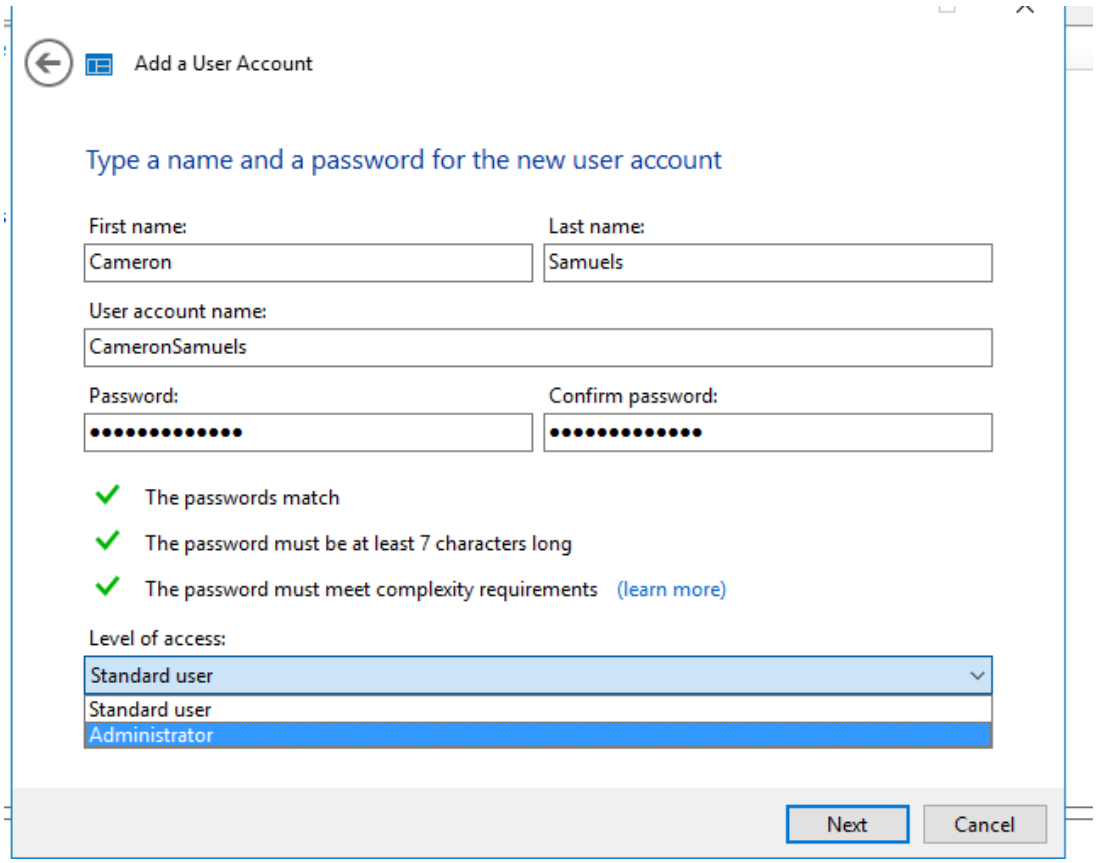


Figure 21 - Creating user accounts in Active Directory.

This process is repeated until all user accounts shown in the organisational user diagram are created. Standard options for settings such as shared access and read or read-write access is selected throughout this process, as this investigation does not go deeper into Windows Server functionality but rather focuses on identity features.

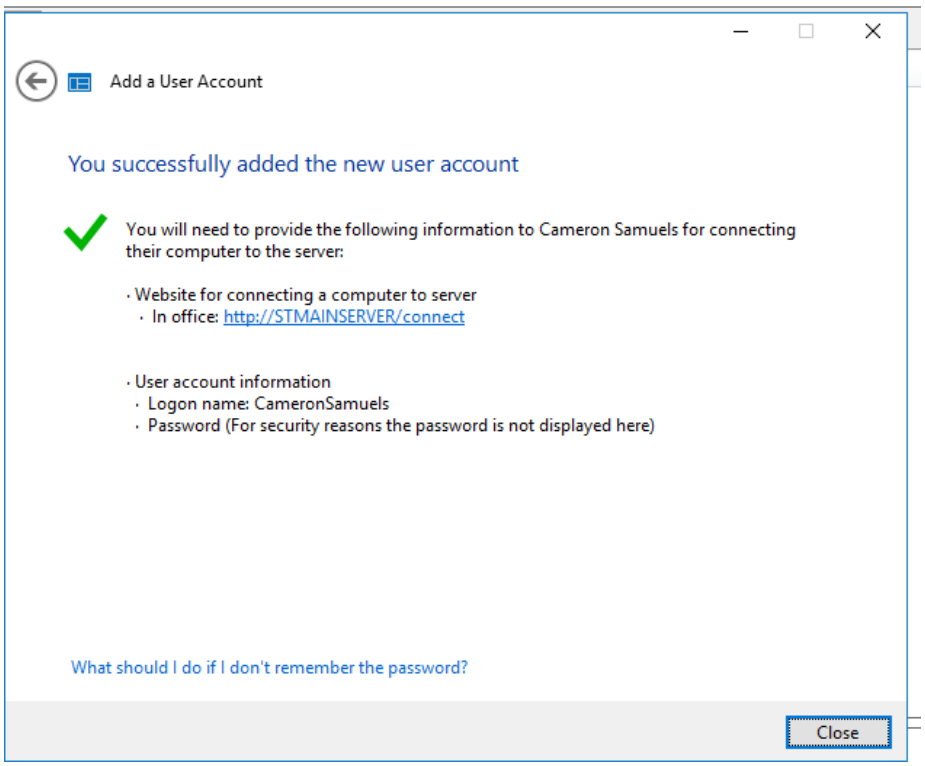


Figure 22 - Account creation completion in Active Directory.

Once all user identities are created in the Active Directory deployment, the Cameron Samuels user account is elevated to Administrator as he is an IT Administrator working under Selena Bass and the Matilda Sands account is also elevated to Administrator as they are the proprietor of the company.

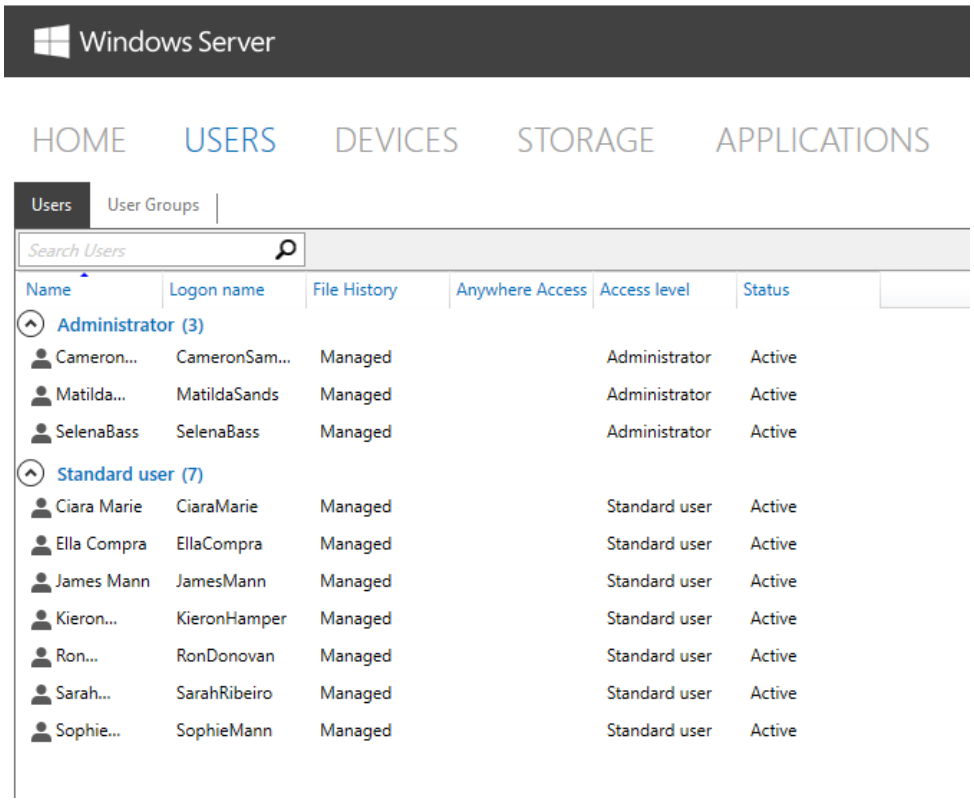


Figure 23 - Users section of Active Directory showing user accounts already created.

Further user groups and administrator types are out of the scope of this investigation, as this investigation focuses on the overall feature and cost benefits rather than specific functionalities within Windows Server.

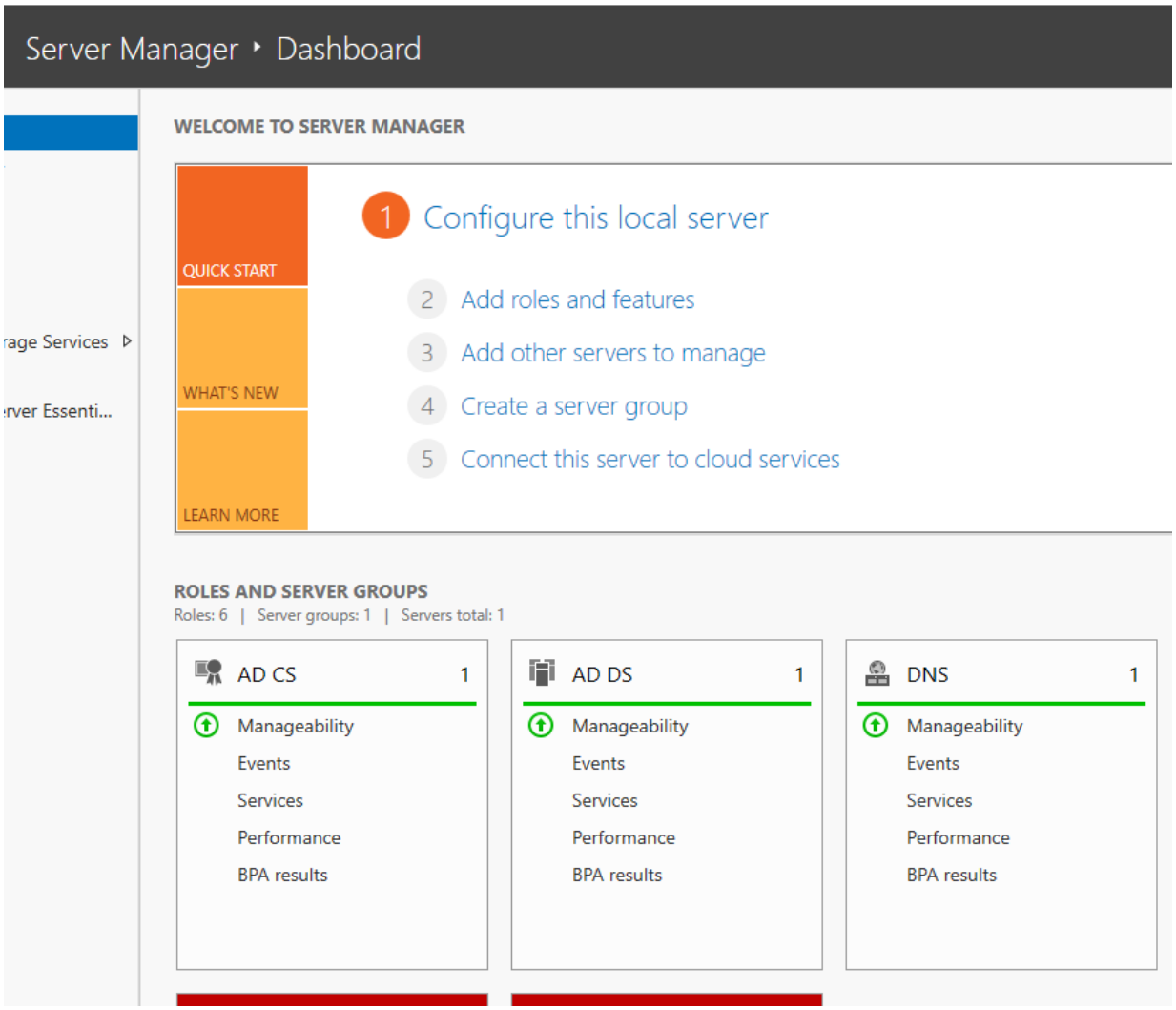
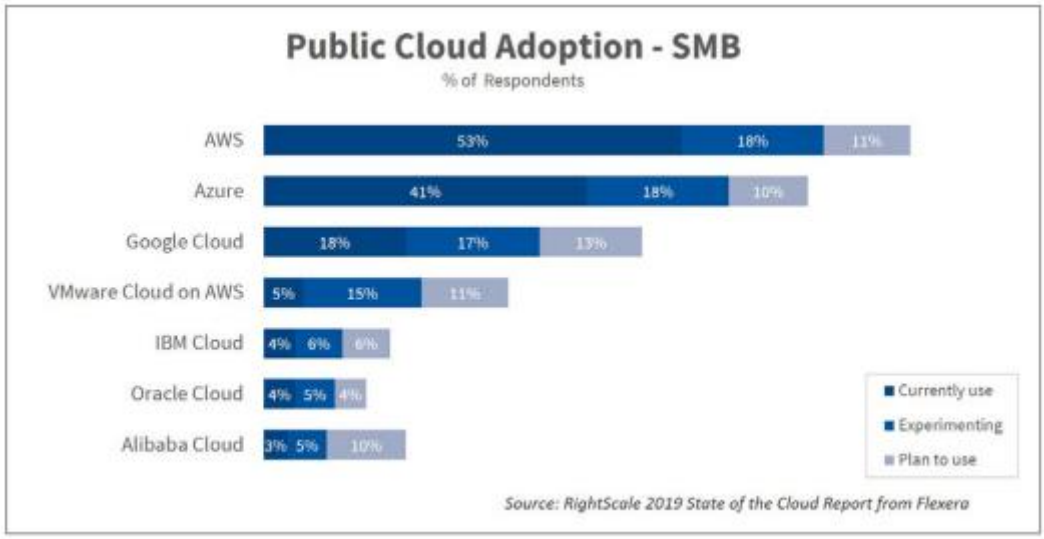


Figure 24 - Server Manager dashboard in Active Directory showing various working components.

Finally, the installation is shown running Active Directory – Directory Services, which is the equivalent of what is run in the Cloud.

Chapter 11 - Appendices



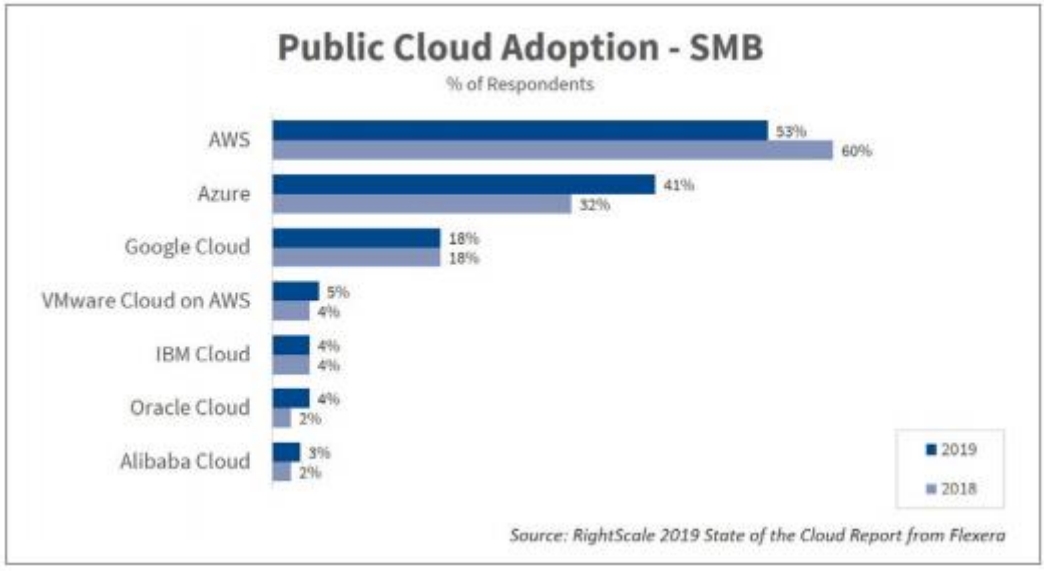


Figure 25 - Public Cloud Adoption statistics for SMB's by RightScale.

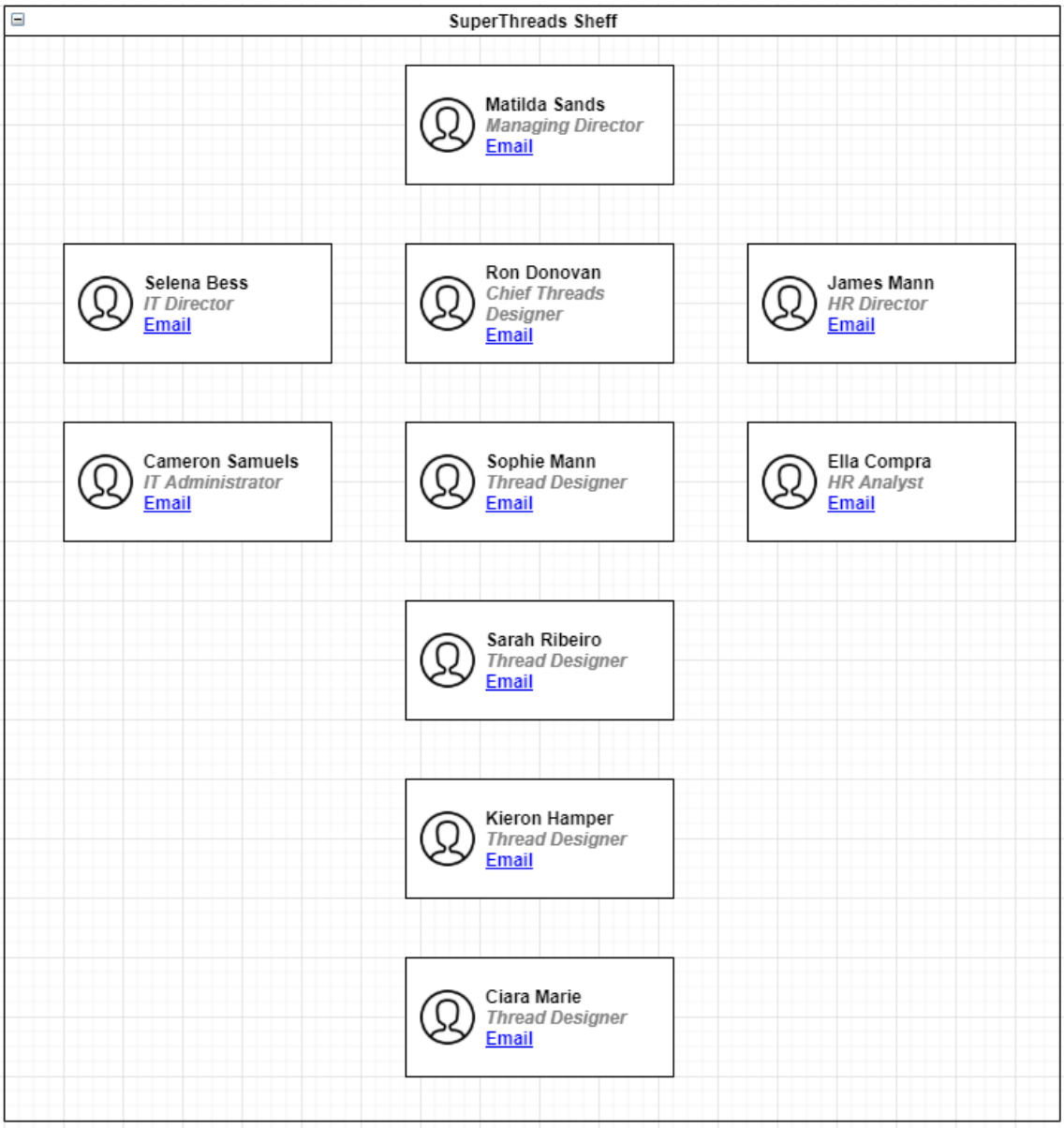


Figure 26 - SuperThreads Sheff's organisational chart showing all individuals in the organisation in a ranked hierarchy.

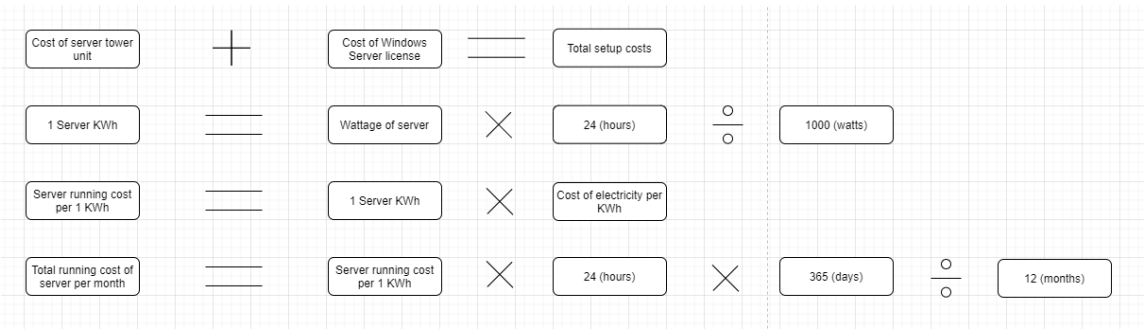


Figure 27 - The mathematical equations used to determine costs and electricity values used in the investigation.

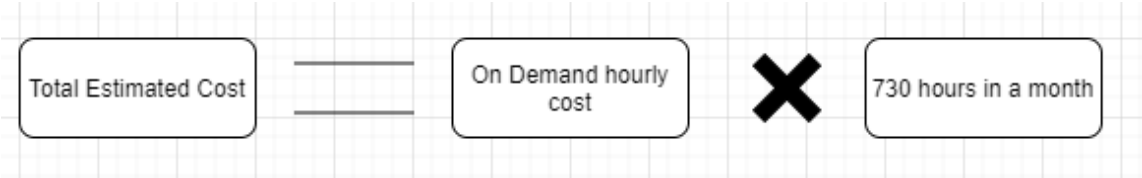


Figure 28 - The mathematical equation used to determine the total estimated cost of running an Active Directory server on Microsoft Azure.

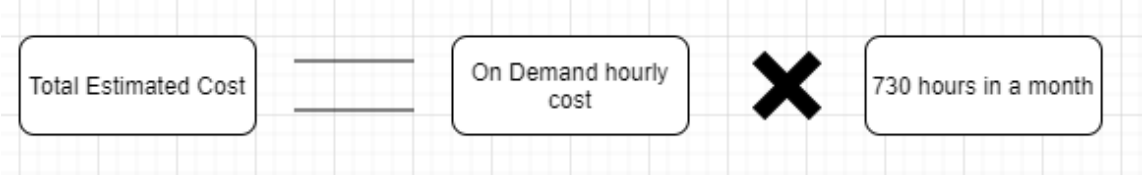


Figure 29 - The mathematical equation used to determine the total estimated cost of running an Active Directory server on Amazon Web Services.

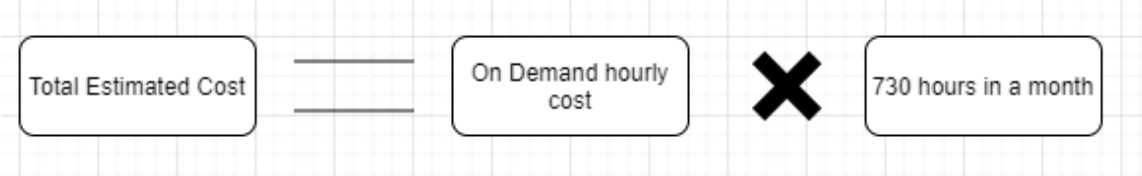


Figure 30 - The mathematical equation used to determine the total estimated cost of running an Active Directory server on Google Cloud Platform.

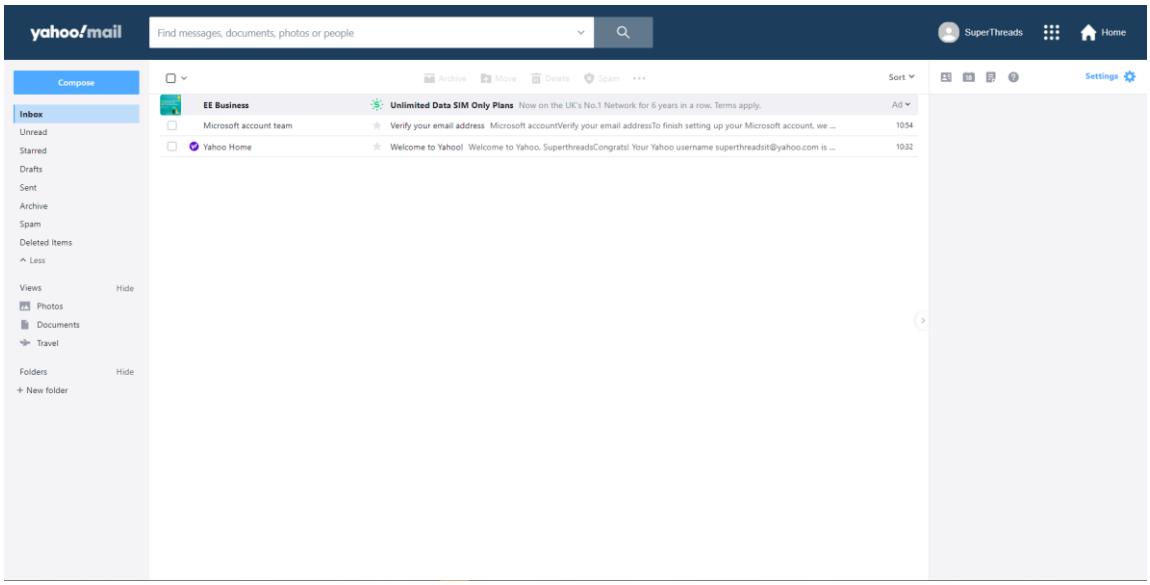


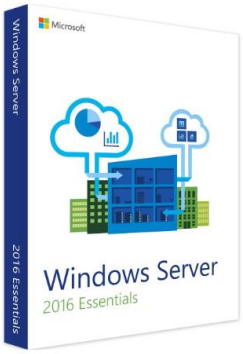
Figure 31 - The Yahoo mail account set up for SuperThreads Sheff as part of the investigation.

servacalc.py

```python
## ServaCalc, a simple calculation tool used to determine the cost of running a physical server over the course of a month by inputting specifications.

try:
    print("Welcome to ServaCalc, the simple server running costs calculation tool. I need to take a few bits of information from you.\n")
    towerCost = float(input("What is the cost of your tower unit in pounds and pence?\n"))
    wattValue = int(input("How many watts does your tower unit use? See the manufacturer's website for details.\n"))
    electricityCost = float(input("Please enter the cost of your electricity in the form of pence per KWh. Check your energy provider for details.\n"))


except ValueError:
    print("Please only enter digits.")


else:

    kwhHour = wattValue * 24 / 1000
    kwhHourCost = float(kwhHour * electricityCost)
    totalcostMonth = float(kwhHourCost * 24 * 365 / 12)

    print(f"Your initial setup costs are £{towerCost:,.2f}".replace('$-', '-$'))
    print(f"Your total monthly server running costs are £{totalcostMonth/100:,.2f}".replace('$-', '-$'))

    ## Calculations.

    # 1000 watts = 1kwh (kilowatt hour) so the full calculation would be the wattValue * hours / 1000 to get the kwh value.
    # For example: 250 watts * 5 hours in a day = 1.25kwh per day.
    # Then to find the total cost, multiply the value of a server KWh by the cost of electricity per KWh.
    # In this project calculation, we will assume the server is running 24 hours a day 365 days a year, as an identity solution should be. We're doing the cost by month, so...
    # The overall math is (cost of KWh * 24 (hours) * 365 (days) / 12 (months).
```

Figure 32 - The code used to build the ServaCalc program in Python for use in the investigation.

lizengo > Microsoft Windows Server > Windows Server > Windows Server 2016



WINDOWS SERVER 2016 ESSENTIALS

(0) Review now

Manufacturer: Microsoft
Item no.: 100065-UK
Status: New
Run time: Permanently valid
Language: All languages

SOFTWARE DOWNLOAD + LICENSE KEY

1 £ 239.99
Prices incl. VAT

ADD TO SHOPPING CART

You will receive 239 bonus points

Figure 33 - Lizengo webpage showing the cost of a Windows Server 2016 Essentials license for use in the investigation.

Power	1x fixed 80 PLUS power supply up to Platinum efficiency; 250W or 400W

Figure 34 - The power figure in Watts for the server used in the investigation from serversdirect.

Last updated · 21 April, 21:56 BST

GBP £ - Pound Sterling	USD $ - US Dollar
919.96	1,130.72

Day | Week | Month | Year | 5 Year | Max

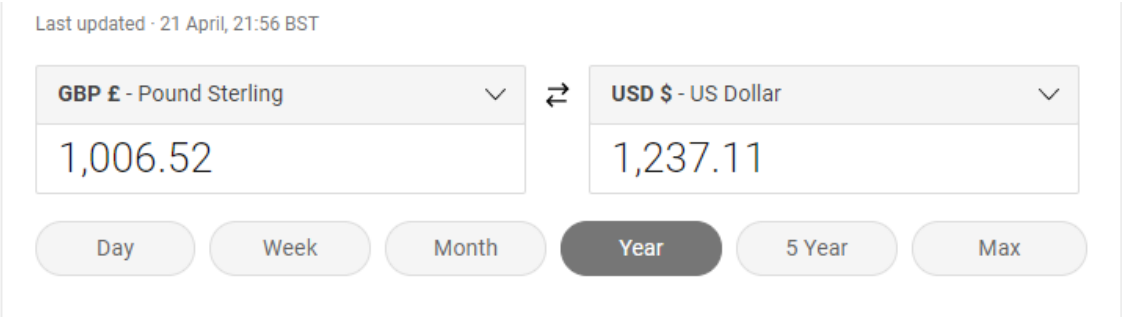


Figure 35 - Spot rate conversion between GBP and USD.

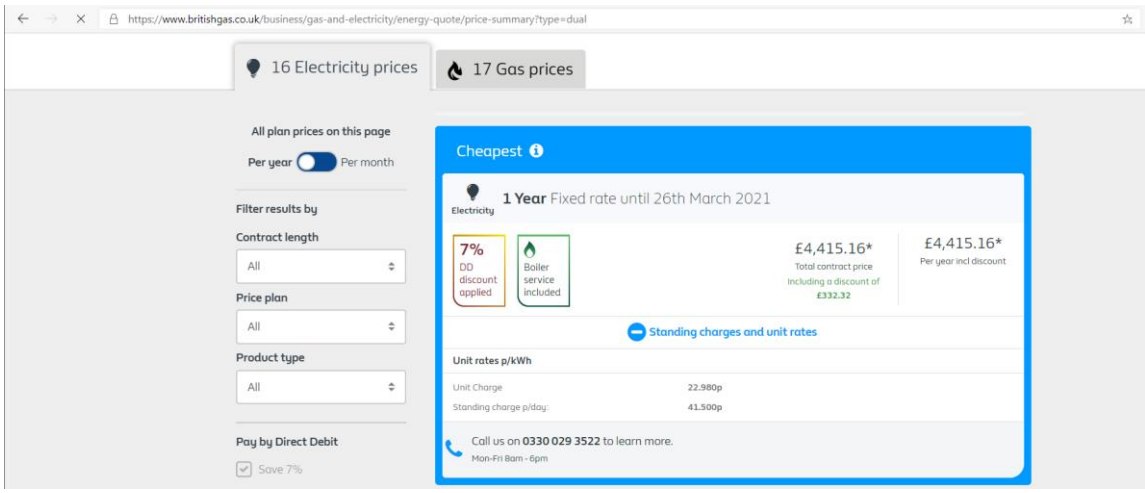
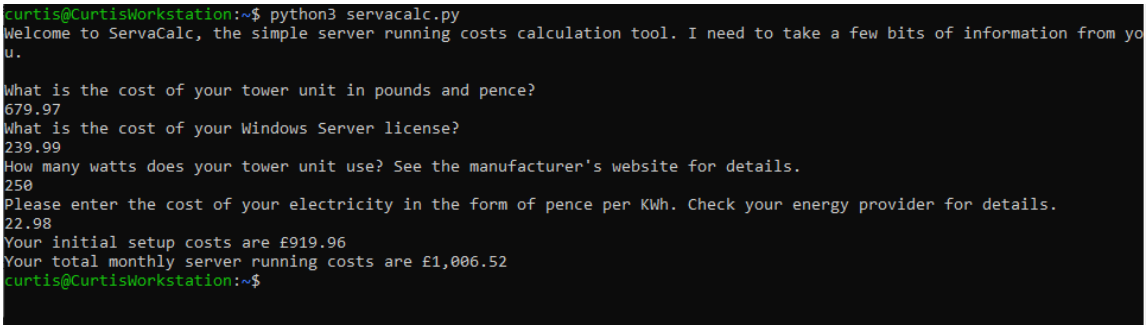


Figure 36 - Electricity prices for the postcode in which SuperThreads Sheff is based (Sheffield, UK).



```
curtis@CurtisWorkstation:~$ python3 servacalc.py
Welcome to ServaCalc, the simple server running costs calculation tool. I need to take a few bits of information from yo
u.

What is the cost of your tower unit in pounds and pence?
679.97
What is the cost of your Windows Server license?
239.99
How many watts does your tower unit use? See the manufacturer's website for details.
250
Please enter the cost of your electricity in the form of pence per KWh. Check your energy provider for details.
22.98
Your initial setup costs are £919.96
Your total monthly server running costs are £1,006.52
curtis@CurtisWorkstation:~$
```

Figure 37 - The ServaCalc python program running from the command line, displaying costs for the investigation.

Your Estimate

Estimate total: US$74.51

Virtual Machines — 1 B2MS (2 vCPU(s), 8 GB RAM) x 730 Hours; Windows – ... — US$74.51

Virtual Machines

REGION: UK West

OPERATING SYSTEM: Windows

TYPE: (OS only)

TIER: Standard

INSTANCE: B2MS: 2 vCPU(s), 8 GB RAM, 16 GB Temporary storage, US$0.102/hour

Clone

Delete

More info

Pricing details

Product details

Documentation

Billing Option

Save up to 72% on pay-as-you-go prices with 1-year or 3-year Reserved Virtual Machine Instances. Reserved Instances are great for applications with steady-state usage and applications that require reserved capacity. Learn more about Reserved VM Instances pricing.

- Pay as you go
- 1 year reserved (~38% savings)
- 3 year reserved (~57% savings)

Save up to 40% with Windows Server Licences you already own. Learn more about Azure Hybrid Benefit to save compute costs.

Benefit to save compute costs.

1 Virtual machines × 730 Hours = US$74.46 Per month

Virtual Machines are billed per second. Learn more about Virtual Machines pricing.

Managed Disks

TIER: Standard HDD

DISK SIZE: S40: 2048 GiB, US$90.101/month

ADD SNAPSHOT

1 Disks × US$90.10 Per month = US$90.10

Storage transactions

100 Transaction units (10,000 transactions) × US$0.0005 Per unit = US$0.05

Sub-total US$164.61

Support

SUPPORT: Included — US$0.00

Support

SUPPORT:

Included

US$0.00

Programmes and Offers

LICENSING PROGRAMME:

Microsoft Online Services Agreement

SHOW DEV/TEST PRICING

Estimated monthly cost

US$164.61

Export Save Share

US Dollar ($)

Prices are estimates and are not intended as actual price quotes. Actual prices may vary depending on the date of purchase, currency of payment and type of agreement that you enter into with Microsoft. Contact a Microsoft sales representative for additional information on pricing.

Purchase options

Figure 38 - Screenshots of cost calculations using the Azure cost calculator.

aws pricing calculator

Feedback English Contact Sales

AWS Pricing Calculator > My Estimate > Add service

Step 1 Select service

Step 2 Configure Amazon EC2

Configure Amazon EC2 Info

Region

EU (London)

Quick estimate

Choose this option for fast and easy route to a ballpark estimate based on minimum requirements or a specific instance search. The estimate assumes consistent utilization.

Advance estimate

Choose this option for a more detailed estimate that accounts for workload, data transfer costs, additional storage options, and other, less common instance requirements. For example, you know that you get a lot of traffic on Mondays but not much traffic throughout the rest of the week, and you want an estimate that takes this workload into account.

EC2 instance specifications Info

Operating system

Choose which operating system you'd like to run Amazon EC2 instances on.

Windows Server

Instance type

Search by name or enter the requirement to find the lowest cost instance for your needs.

Enter minimum requirements for each instance:

Search instances by name:

vCPUs 1 Remove

Memory (GiB) 8 Remove

Add requirement

aws pricing calculator

Feedback English Contact Sales

Memory (GiB) 8 Remove

Add requirement

Based on your inputs, this is the lowest-cost EC2 instance:

t3a.large

On-Demand hourly cost	vCPUs	GPUs
0.1126	2	NA
1YR Std reserved hourly cost	Memory (GiB)	Network performance
0.0876	8 GiB	Low to Moderate

Quantity

Enter the number of Amazon EC2 instances that you need.

1

Pricing strategy Info

Pricing model	Reservation term	Payment options
EC2 Instance Savings Plans	1 Year	No Upfront
Compute Savings Plans	3 Year	Partial Upfront
Standard Reserved Instances		Full Upfront
Convertible Reserved Instances		
On-Demand Instances		

Show calculations

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Terms of Use Privacy Policy

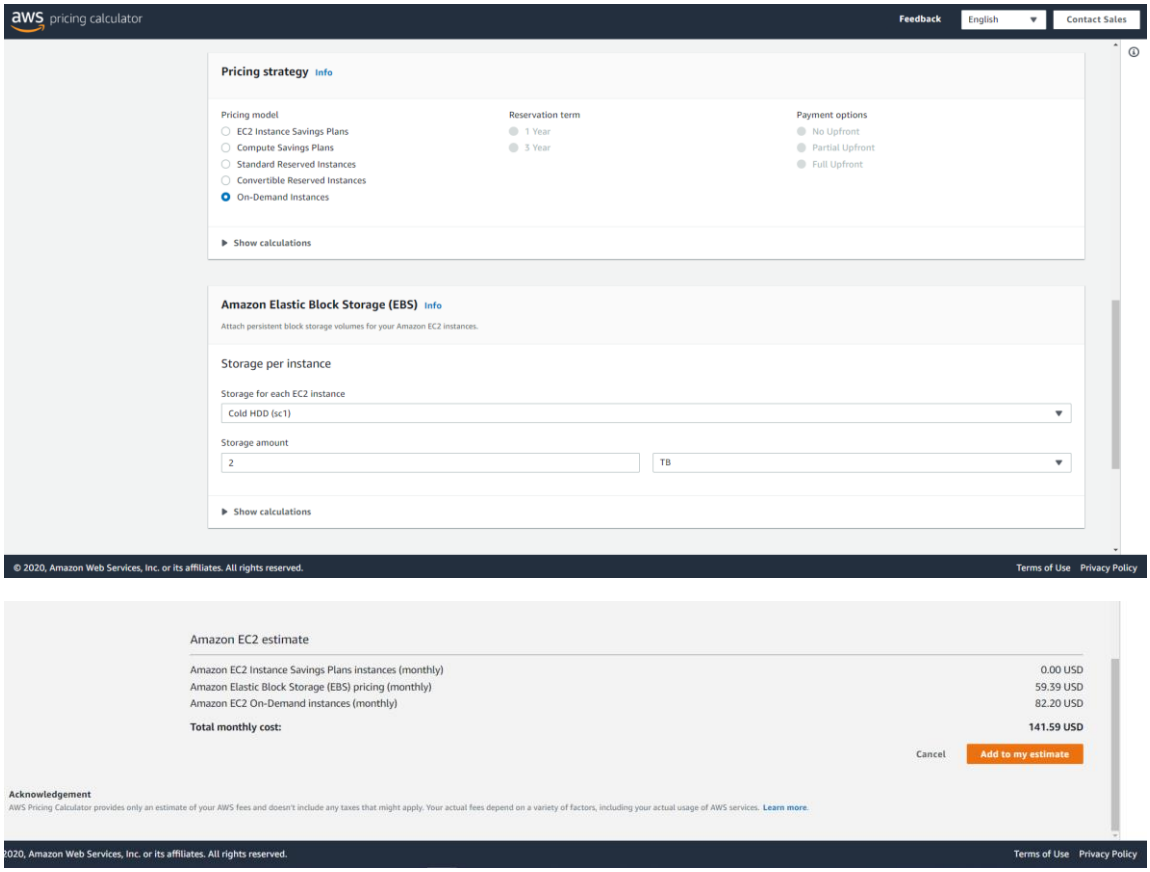
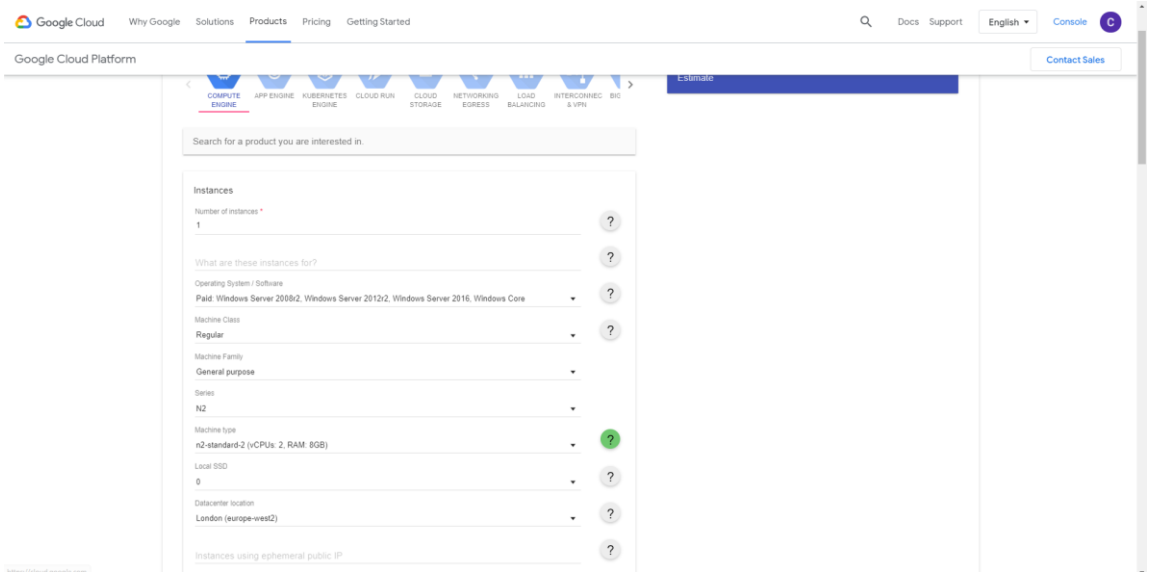
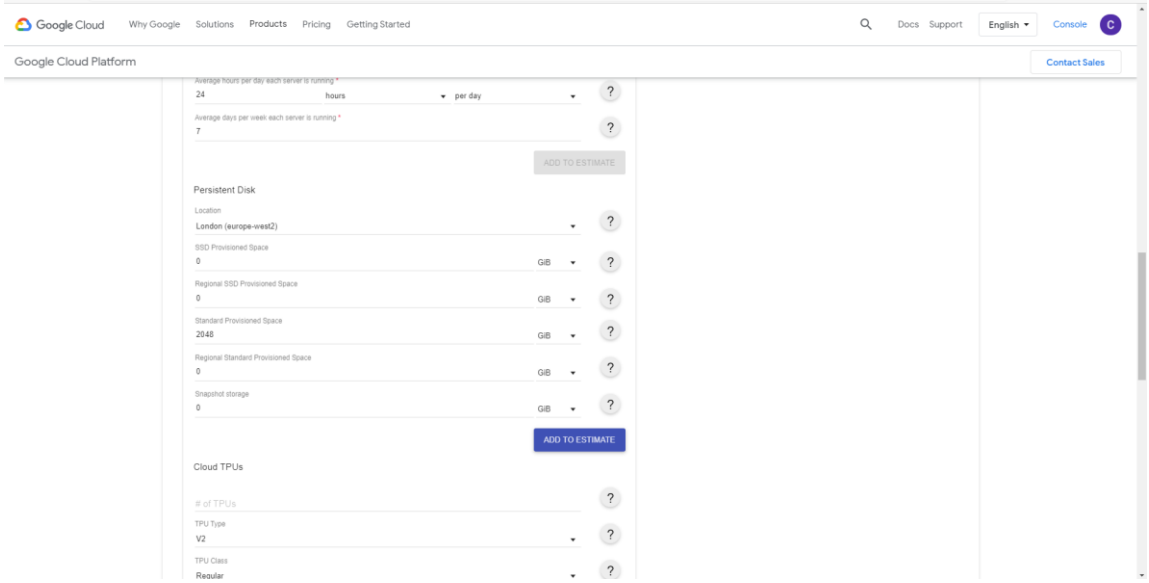


Figure 39 - Screenshots of costs using the Amazon Web Services cost calculator.




Google Cloud
Why Google
Solutions
Products
Pricing
Getting Started
Docs
Support
English
Console
Google Cloud Platform
Contact Sales
Average hours per day each server is running *
24
hours
per day
Average days per week each server is running *
7
ADD TO ESTIMATE
Persistent Disk
Location
London (europe-west2)
SSD Provisioned Space
0
GiB
Regional SSD Provisioned Space
0
GiB
Standard Provisioned Space
2048
GiB
Regional Standard Provisioned Space
0
GiB
Snapshot storage
0
GiB
ADD TO ESTIMATE
Cloud TPUs
# of TPUs
TPU Type
V2
TPU Class
Regular

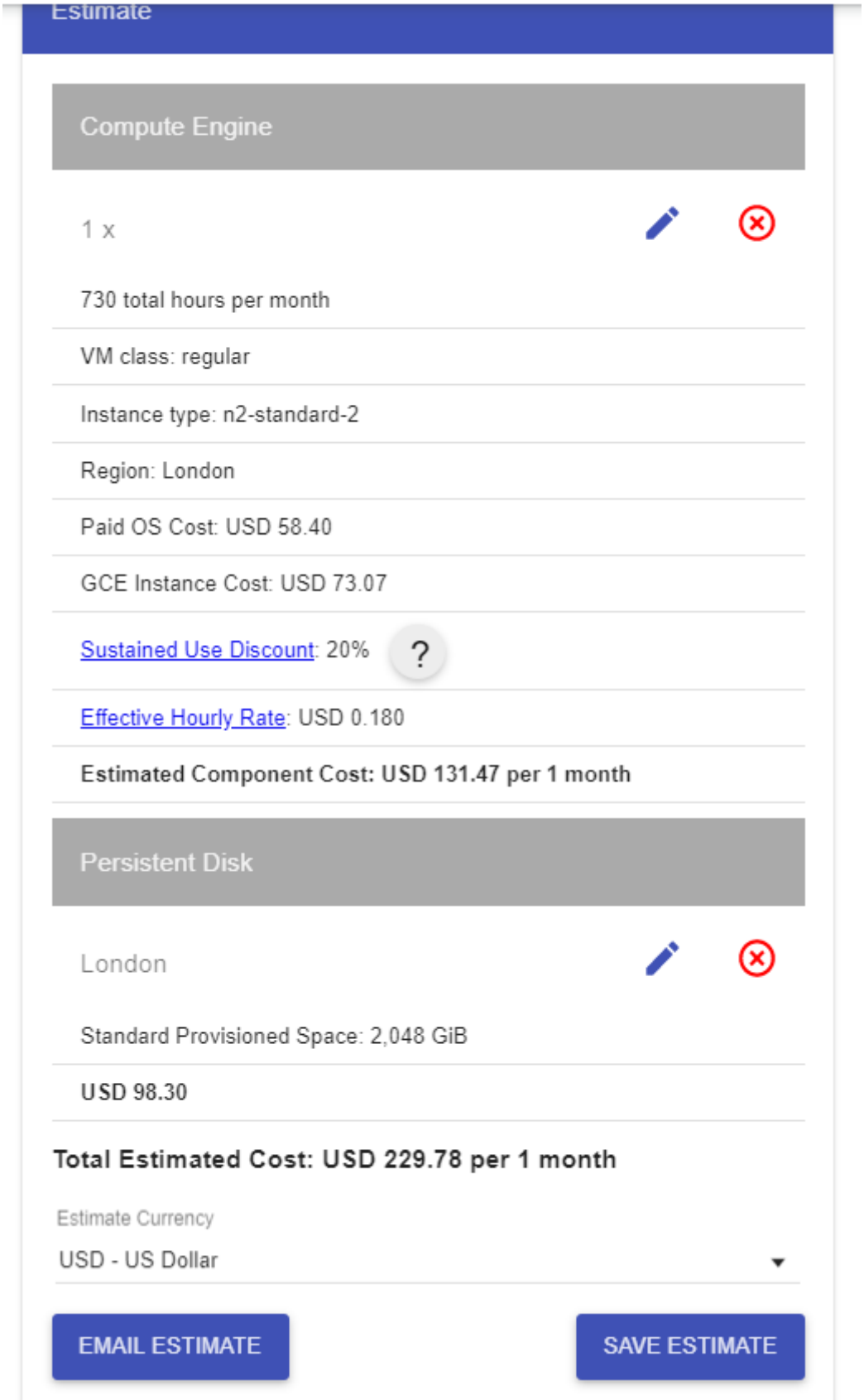


Figure 40 - Screenshots of cost using the Google Cloud Platform cost calculator.

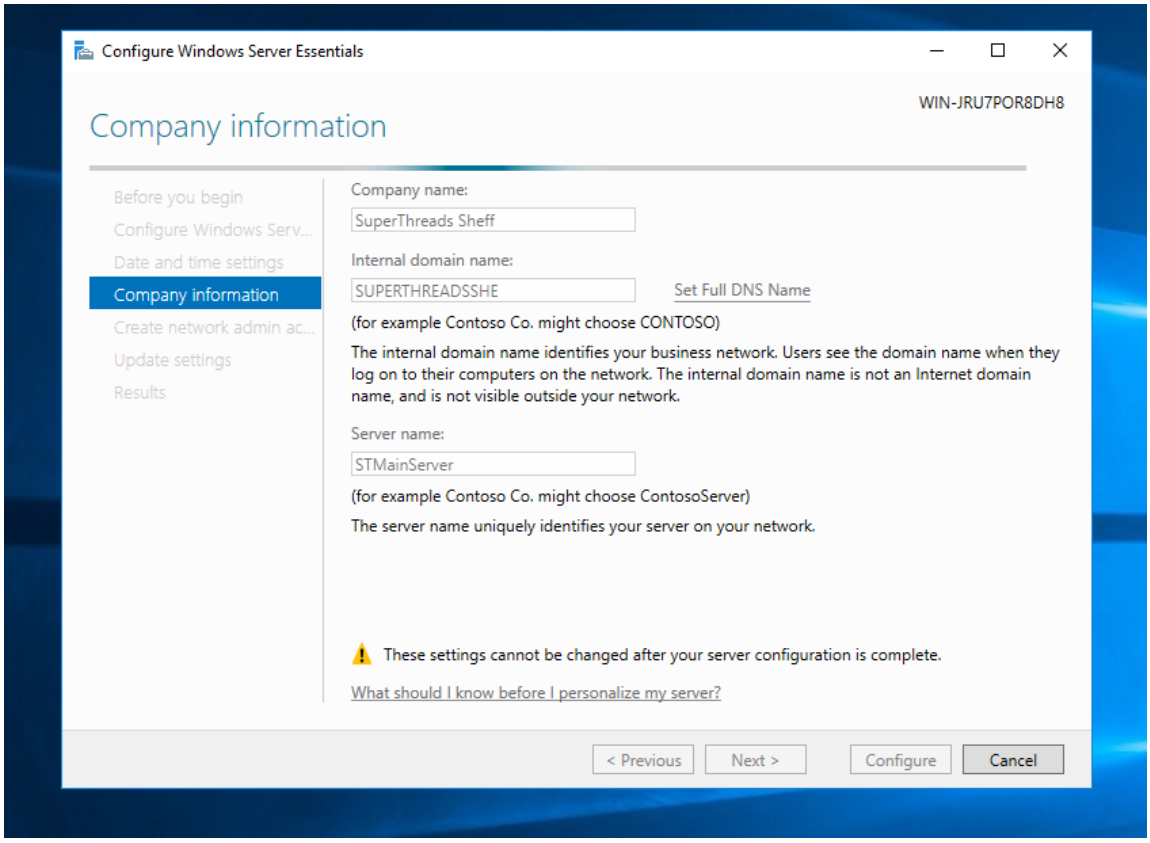


Figure 41 - Setting up Company information in Windows Server Essentials.

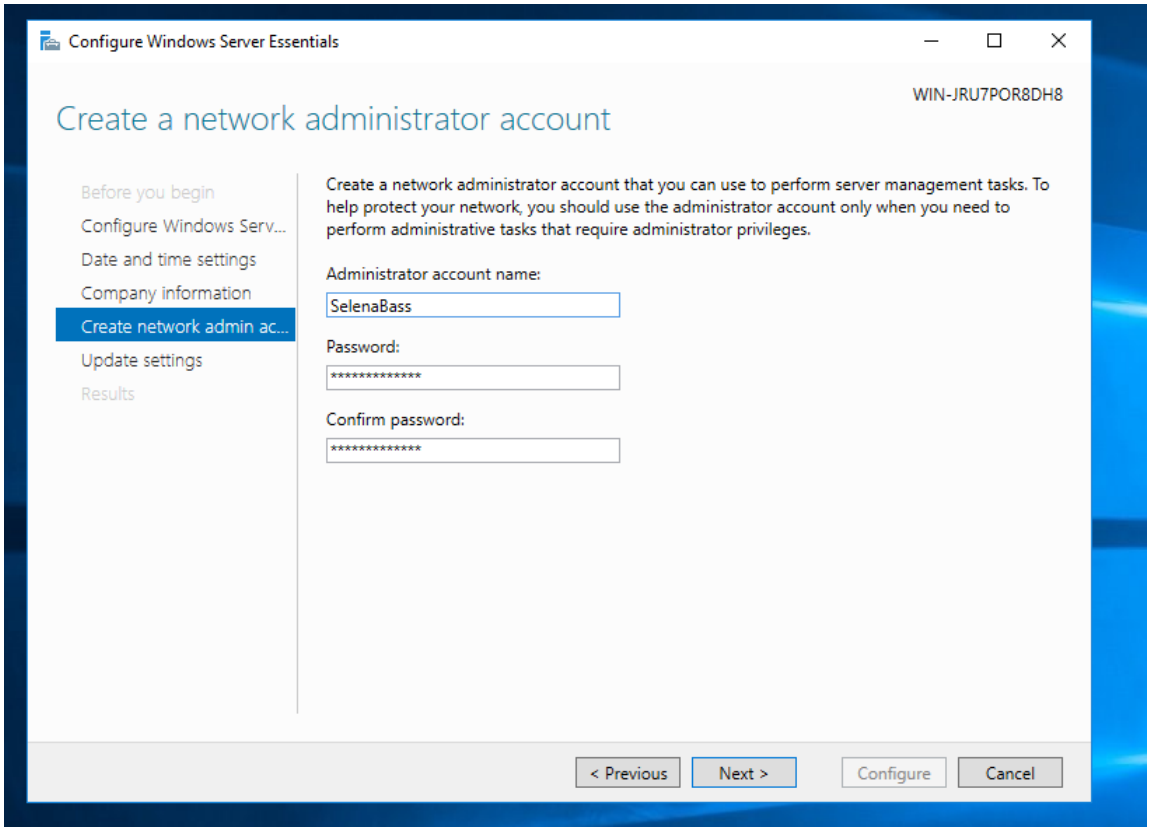


Figure 42 - Creating an administrator account in Active Directory.

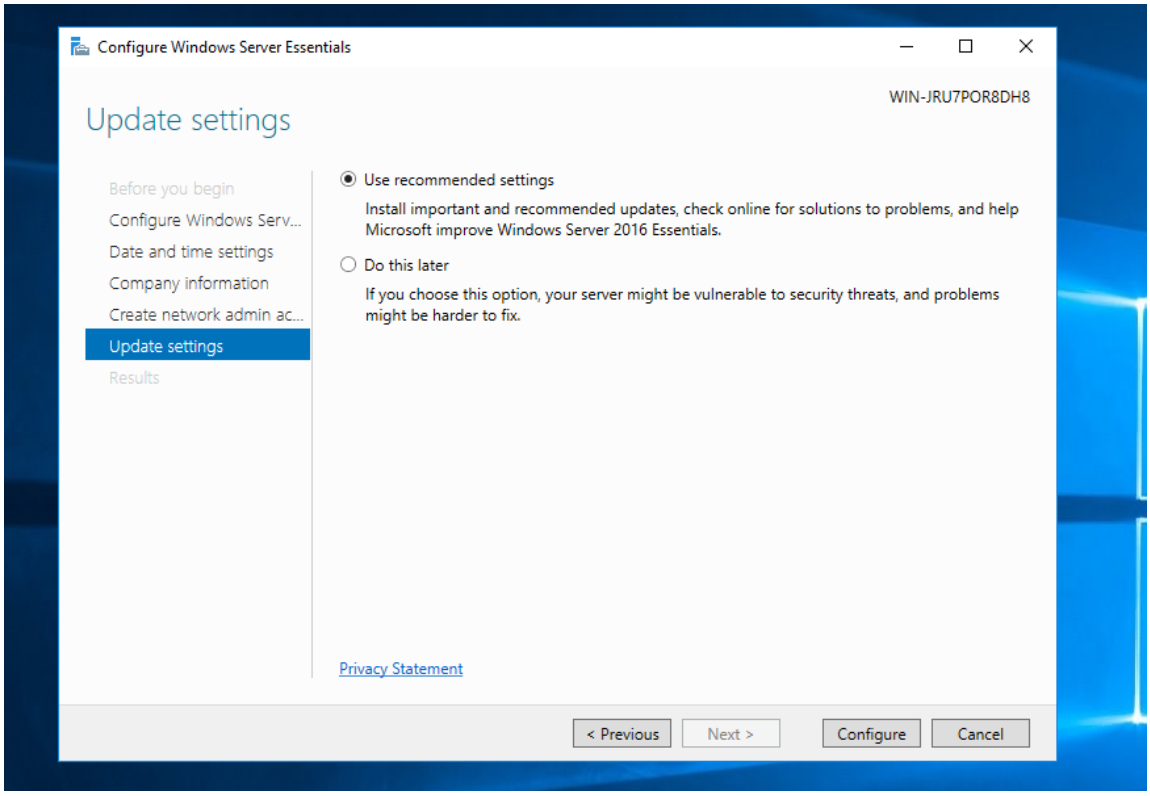


Figure 43 - Selecting settings to use in Windows Server Essentials.

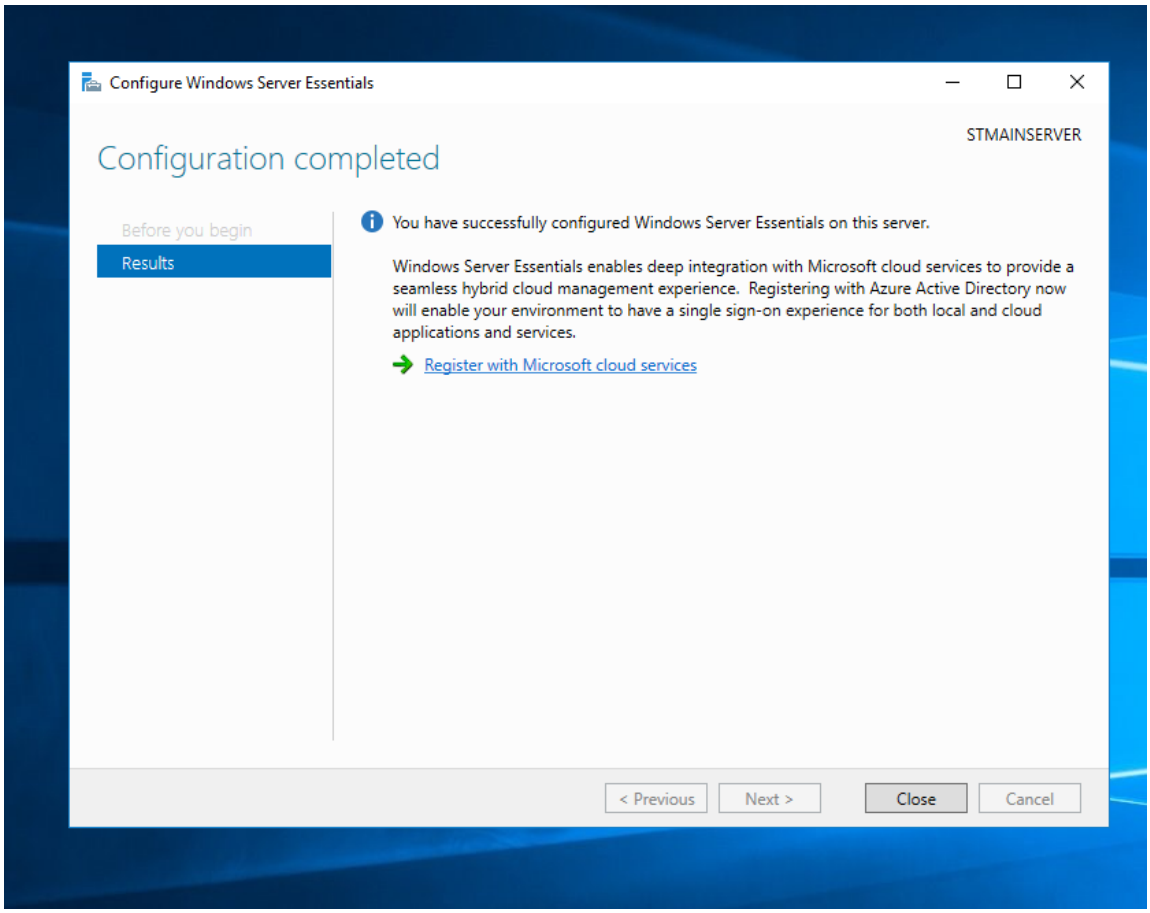


Figure 44 - Windows Server Essentials completing configuration.

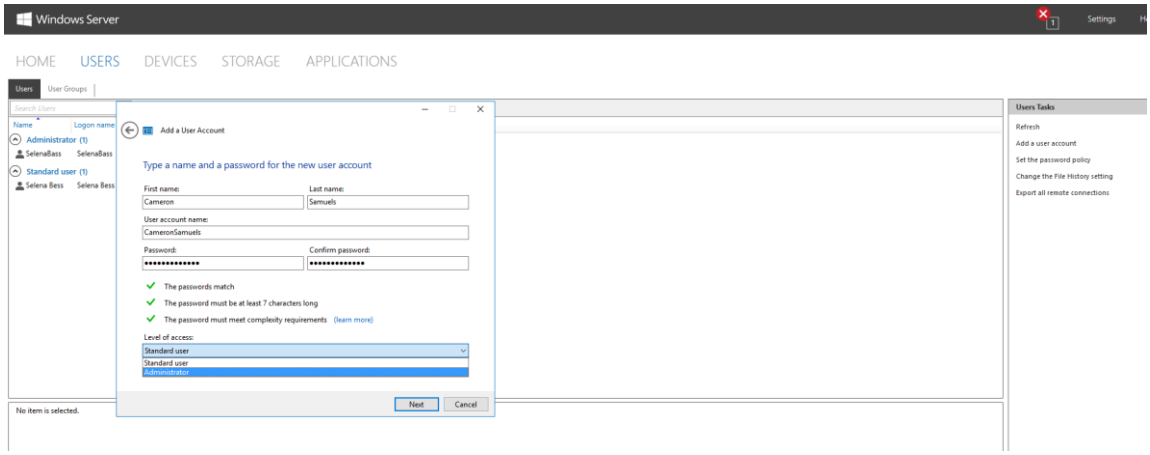


Figure 45 - Creating user accounts in Active Directory.

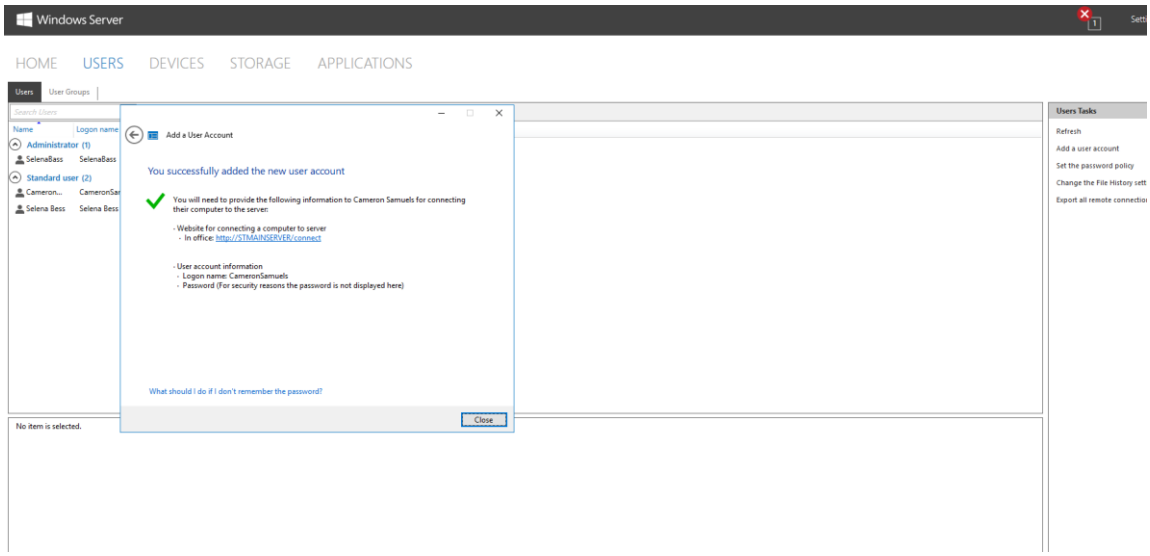


Figure 46 - Account creation completion in Active Directory.

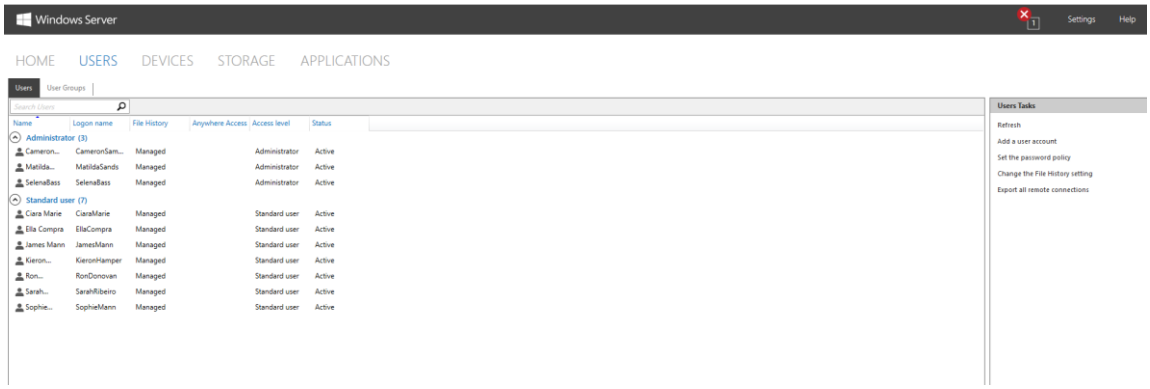


Figure 47 - Users section of Active Directory showing user accounts already created.

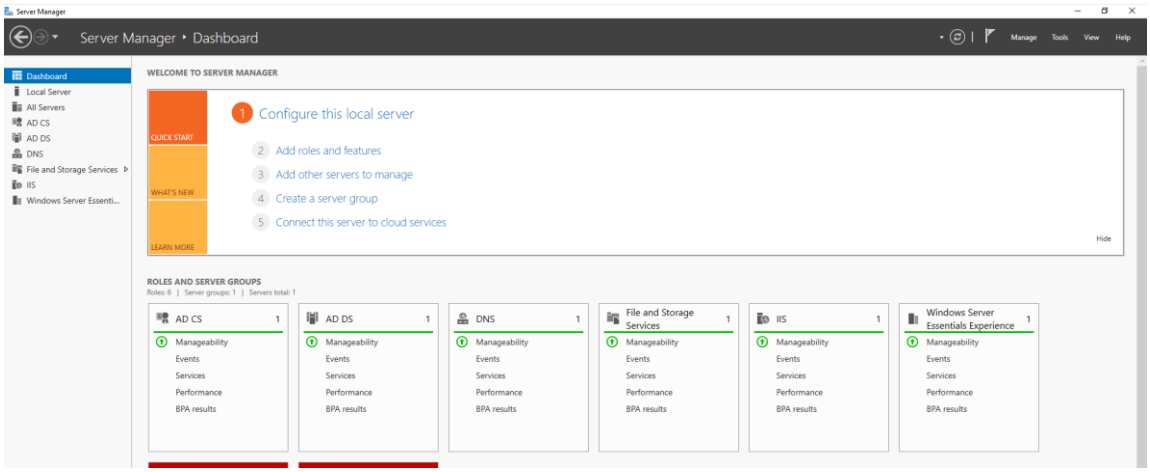


Figure 48 - Server Manager dashboard in Active Directory showing various working components.

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